

RUNTIME ENGINEERS • STRATEGY AUTHORS

RUNTIME MANUAL

Evaluator, series model, builtins, strategy, compiler

Open-source Pine Script evaluation framework



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OVERVIEW

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title: "Runtime" description: "Bar-loop evaluator — series caps, incremental TA, alerts, fill export, foreign request.security → na, and interpret ↔ compile parity."

RUNTIME

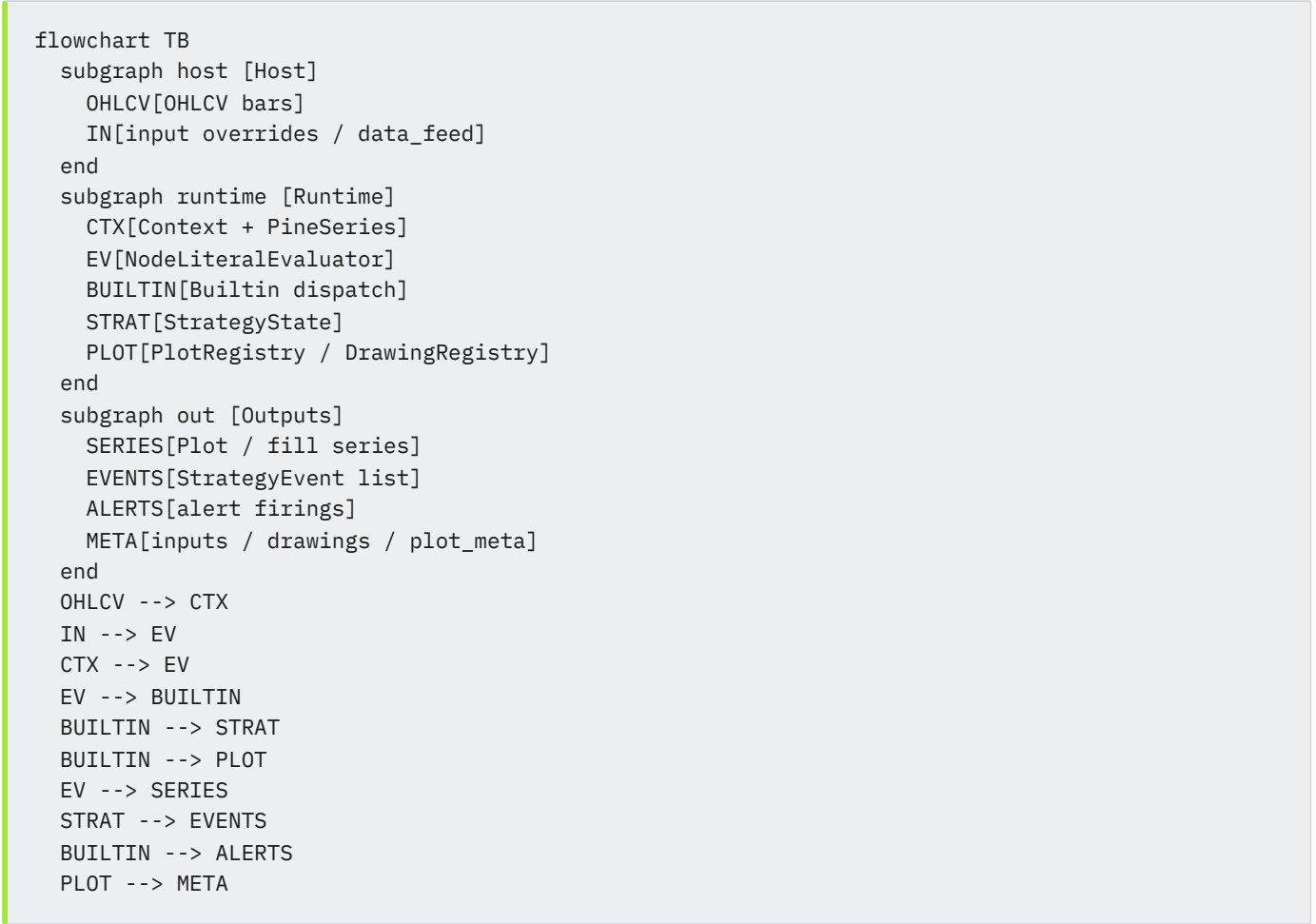
Abstract

PYNE's runtime is the deterministic **bar-loop** that turns a parsed ASDL AST into series values, drawing side-effects, strategy events, and **alerts**. It is deliberately host-agnostic: the same visitor mixins power the library API, the Flask Pro API (`Runtime.run`), browser Pyodide, and edge workers. A second path—source-to-source compilation into a Numba or object-mode bar loop—trades interpretive flexibility for throughput on long histories; Pro API defaults to `mode=auto`.



This section documents **semantics**, not the chart. AXIS consumes plots / `fill` / drawings; hosts export alerts (optional L2 webhooks); HOOX consumes strategy events. None of those hosts is required to evaluate a script.

Conceptual model



Execution modes share the same parse tree (Pro API / `Runtime` also expose `auto`):

Mode	Entry	Bar loop	Typical use
Interpret	<code>NodeLiteralEvaluator.visit(tree)</code> per bar	Host updates <code>PineSeries</code> / context, re-visits AST	Full language surface, strategy, drawings, alerts, <code>request.*</code>
Compile	<code>pynescrypt.compiler.engine.compile_script</code>	Generated for <code>__bar_idx</code> in <code>range(n)</code> over OHLCV + <code>time_arr</code>	Numeric indicators (Numba) or object-mode subset
Auto	<code>Runtime.run(..., mode="auto")</code>	Compile when safe; interpret on fallback	Pro API default — prefer throughput without hard-failing exotic scripts

Compile inputs are contiguous arrays: `open_arr ... vol_arr` plus `time_arr` (bar-open Unix ms; synthetic `bar_index * 60_000` when the host omits times) so `time` / calendar builtins match the interpret `time` series. Disk IR cache + product warm-compile (prewarm API/CLI) cut cold JIT latency. See [Compiler overview](#) and [Interpret ↔ compile parity](#).



request.security (landed): interpret resamples same-symbol simple OHLCV and allowlisted `ta.sma` / `ema` / `rsi` / `atr` on a **coarser** TF (last **completed** HTF bucket only). Compile passthrough is same-symbol simple OHLCV only. Foreign symbols and complex / nested expressions resolve to `na` on both hosts (no inventing chart close as foreign fundamentals). Policy tags land on `result["meta"]["request_security"]`. Real multi-symbol feeds remain adapter work (roadmap **B1**).

Series caps / incremental TA: host list trim defaults `on` (`PYNE_SERIES_CAP`); bar-mode TA hot path defaults `on` (`PYNE_TA_INCREMENTAL`). Drawing GC honors `max_*_count` on the interpret registry; `fill` exports series keys + `plot_meta` for AXIS bands.

Interpret hot path (0.3.10 / Round 9): the AST walker inlines `Assign / Expr(Call)` after the first bar (function/type/import decls stay locked), skips unused derived OHLCV series (`hl2` / `hlc3` / `ohl4` / `tr` unless named, `input.source`, or `ta.vwap`), and reuses plot cells by call-site. Same-machine bench @ 2000 bars vs 0.3.9: minimal **2.7×**, `ta.sma` **2.0×**, `ta_combo` **1.45×**. `PYNE_SERIES_RING` stays `off`.

Host flags and knobs

Knob	Default	Effect
<code>PYNE_SERIES_CAP</code>	<code>on</code>	Trim chronological <code>current_series</code> lists (<code>maxBarsBack</code> / 256)
<code>PYNE_TA_INCREMENTAL</code>	<code>on</code>	Call-site incremental <code>ta.*</code> in bar mode
<code>PYNE_SERIES_RING</code>	<code>off</code>	Chronological ring lookback; skip dual list write when on
<code>PYNE_LIGHT_PLOTS</code>	<code>off</code>	Skip plot columns + input meta (corpus OK/fail only)
<code>PYNE_RUNTIME_MODE</code>	<code>interpret</code>	Default when <code>Runtime.run(mode=)</code> is omitted (<code>auto</code> is the Pro API body default)
<code>timeout_seconds=</code>	<code>None</code>	Interpret wall-clock budget; checked every 32 bars → <code>timed_out</code> + <code>error_kind=runtime</code>
<code>libraries=</code>	<code>[]</code>	<code>{namespace, name, version, source}</code> registered before <code>import</code> (auto forwards into interpret fallback)
<code>realtime_*</code>	<code>historical</code>	Interpret-only <code>varip</code> / forming-bar simulation (<code>realtime_last_bar</code> / <code>ticks</code> / <code>bars</code> / <code>from_bar</code>)
<code>Runtime.create_session</code>	<code>interpret</code>	Persistent <code>InterpretSession</code> (0.6.1): delta <code>append_bars</code> , forming-bar <code>update_last_bar</code> ; compile has no resumable state



Interface surface

Concern	Start here
Series history, <code>[]</code> , <code>var</code> / <code>varip</code> , <code>PYNE_SERIES_CAP</code> , <code>na</code>	Series & history
Expressions, statements, control flow	Expressions & statements
Builtin namespaces (<code>ta.*</code> , <code>strategy.*</code> , ...)	Builtins hub
Drawing/ <code>plot</code> / <code>fill</code> / <code>max*_count</code> GC	Drawing & plotting
<code>request.*</code> / <code>input.*</code> (foreign → <code>na</code>)	request / input
Library <code>export</code> / <code>import</code>	Libraries
<code>StrategyEvent</code> parity contract	Events
<code>alert()</code> / webhooks	Alerts
Numba / object-mode compiler	Compiler overview
Interpret vs compile plot parity	Parity testing
Incremental interpret sessions (<code>create_session</code>)	Evaluate scripts

Library callers typically use `NodeLiteralEvaluator` or the package `pynescrypt.runtime.Runtime` (Pro API re-export) rather than wiring the bar loop by hand:

```
from pynescrypt.ast.evaluator import NodeLiteralEvaluator
from pynescrypt.runtime import Runtime # package SoT (0.3.4+)

ev = NodeLiteralEvaluator(context={"close": [1.0, 2.0, 3.0], "bar_index": 2})
result = ev.evaluate_script('//@version=6\nindicator("x")\nplot(close)')

# Full bar loop (preferred host API):
# Runtime(symbol="CHART").run(
#     source, ohlcv_data=bars, mode="auto",
#     timeout_seconds=None, libraries=None,
# )
# Incremental interpret (0.6.1):
# sess = Runtime(symbol="CHART").create_session(source, bars)
# sess.append_bars(new_closed_bars)
# sess.update_last_bar(forming_bar)
```

Hosts that own OHLCV (Pro API, workers) use the package host: push bar → fill pending orders → `visit(tree)` → drain events → collect plots. **SoT:** `src/pynescrypt/runtime/` (`host.py`, `series.py`, `evaluator.py`).
`backend.runtime` / `backend.series` / `backend.evaluator` are identity re-export shims for the Pro API.



Internals

Path	Role
src/pynescrypt/runtime/	Package Runtime SoT — host, series, CustomEvaluator
src/pynescrypt/ast/evaluator/	Mixin evaluators (base, names, expressions, statements, libraries, events)
src/pynescrypt/ast/evaluator/builtins/	Builtin dispatch tables
src/pynescrypt/compiler/	CompilerVisitor, Numba builtins, strategy broker, engine façade
backend/runtime.py	Compat re-export (sys.modules alias) of pynescrypt.runtime.host
tests/test_evaluator.py, tests/test_parity.py, tests/test_strategy_*.py, tests/test_first_party_ta_goldens.py	Semantic oracles + dual-host TA goldens

Composition of the full interpreter:

```
BaseEvaluator
+ LiteralEvaluator
+ NameEvaluator
+ ExpressionEvaluator
+ StatementEvaluator
+ BuiltinEvaluator
→ NodeLiteralEvaluator
```

BuiltinEvaluator itself aggregates mixins (TechnicalAnalysisMixin, StrategyBuiltinsMixin, PlottingFunctionsMixin, ...) into one dispatch map built at construction time.

Invariants & edge cases

- Bar re-entrancy.** The AST is visited once per bar. Function/type definitions lock after bar 0 (_pine_defs_locked) so multi-dispatch tables do not grow ($O(\text{bars}^2)$).
- Order of broker steps.** Pending limit/stop fills run **before** script body evaluation on each bar (interpreter and compile object-mode agree). Pending-fill averaging when pyramiding ≤ 0 matches both brokers (**F2**).
- na is None.** Unresolved missing data, OOB history, and arithmetic with missing operands produce Python None, not IEEE NaN—except on the Numba path, which uses np.nan as the array sentinel. Cross-mode plot compares must normalize (scripts/compare_interp_compile.py).
- Series caps.** Unbounded current_series growth is capped (PYNE_SERIES_CAP default ON, maxBarsBack / _SERIES_MAX) so long histories stay memory-bounded (**T1**).
- Incremental TA.** Hot-path kernels (MAS, oscillators, bb / kama / cmo / stochrsi, volume obv / wad / cmf / klinger / nvi / pvi, ...) use call-site incremental updates in bar mode (PYNE_TA_INCREMENTAL=0 to disable).
- Strategy state is per-evaluator.** StrategyState lives on the instance (evaluator._strategy_state), never as a process-global singleton—required for concurrent runs and parity tests.
- Drawing/plot registries are side channels.** Visual objects do not participate in pure expression values except where plot() returns a plot id for fill(). Hosts export fill bands for AXIS via series + plot_meta (and



- `__drawings` on compile). Drawing objects honor `max_*_count` GC (package + Pro API + AXIS Pyodide).
8. **Alerts are a host side-channel.** `alert()` / `alertcondition()` firings are not strategy events; dual-host export + L2 webhooks are documented under [Alerts](#).
 9. **Host series are not aliased on assign.** `last = time` (or `open` / `high` / ...) copies the current scalar into a fresh series for `last` —see [Series & history](#).
 10. **Omitted bid/ask stay na.** Host `bid` / `ask` update only when those keys appear on a bar dict; they are not synthesized from close.

Worked example — minimal bar loop mental model

```
for bar_index, bar in ohlcv:
    open/high/low/close.update(bar)
    context[bar_index, time, barstate, ...] = ...
    process_pending_orders(OHLC)    # broker sim
    evaluator.visit(ast)             # script body
    events += drain_events()
    series.append(plot values)
```

Compile mode collapses this into one generated function over full arrays (`open_arr ... vol_arr, time_arr`), with `__bar_idx` standing in for `bar_index`.

Failure modes

Symptom	Likely cause
unexpected type of node: ...	AST node not implemented in any mixin's <code>visit_*</code>
Unknown built-in function: ...	Missing dispatch key or wrong qualified-name resolution
Divergent strategy events vs TV	Partial-fill / OCA / risk settings, or fill timing vs bar
Numba path raises / falls back	Script uses UDT/map/drawing → object mode; or Numba not installed
Plots empty but no error	Host forgot to collect <code>plot_outputs</code> / <code>PlotRegistry</code> after visit; or <code>PYNE_LIGHT_PLOTS=1</code>
<code>timed_out</code> + partial series	<code>timeout_seconds</code> elapsed (checked every 32 interpret bars)

See also

- [Language core](#) — grammar, AST, types before evaluation
- [Alerts](#) — `alert()` engine + L2 webhooks
- [Pro API runtime bridge](#)
- [Interpret ↔ compile parity](#) — `scripts/compare_interp_compile.py`
- [Numerical validation](#)
- [Roadmap](#) — H1/H2/T1/T2/F1/F2/L2/P1p official builtins landed; optional set0x unmeasured
- [pyne-worker](#)
- [AXIS docs](#) — optional AXIS for series/drawings

SERIES AND HISTORY

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title: "Series and history" description: "Pine series model: history operator, na propagation, var/varip persistence, and bar-mode scalars."

SERIES AND HISTORY

Abstract

Pine's core data structure is the **series**: a value that evolves bar by bar, with random access to past values via the history operator `[]`. PYNE implements that model with host-managed series wrappers (`PineSeries` in the Pro API runtime), plain Python lists in unit tests, and flat `numpy` arrays on the compile path. Understanding history indexing and `na` is prerequisite to every builtin and strategy claim.



Conceptual model

```
flowchart LR
    subgraph bar_t [Bar t]
        C0["series[0] current"]
    end
    subgraph bar_tm1 [Bar t-1]
        C1["series[1]"]
    end
    subgraph bar_tm2 [Bar t-2]
        C2["series[2]"]
    end
    C0 -.->|history| C1
    C1 -.->|history| C2
```

Pine indices are **most-recent-first**:

Pine	Meaning	List representation (chronological)
<code>close[0]</code>	Current bar	<code>list[-1]</code>
<code>close[1]</code>	Previous bar	<code>list[-2]</code>
<code>close[n]</code> with $(n \geq \text{len})$	Out of range	<code>na</code> (<code>None</code>)

Negative offsets are **not** Python wraparound; they **soft-fail** to `na` (`None`).

Interface surface

History operator (`visit_Subscript`)

Implemented in `NameEvaluator.visit_Subscript`:

- Float indices coerce to `int` (e.g. `depth / 2`); NaN index $\rightarrow$ `na`.
- Lists map Pine index (i) to Python index $-(i + 1)$.
- Scalars: `x[0]` is `x`; `x[i]` for $(i > 0)$ is `na` (no history buffer).
- Matrices use `[row, col]` (tuple/list of length 2), not series semantics.

Series wrappers

Hosts expose OHLCV as objects with:

- `.current` — scalar for the active bar
- `.history` — most-recent-first buffer (deque or list)

Arithmetic in `ExpressionEvaluator` coerces such wrappers via `_as_scalar_operand` so `close + 1` does not attempt object addition.

Assigning host series (copy, not alias)

Binding a **host** multi-bar series into a **new** name copies the current scalar into a fresh series for that name—it does **not** alias the underlying `open` / `high` / `low` / `close` / `volume` / `time` buffers:

```
last = time           // copy of current time into series 'last'
last := last + 1      // mutates only 'last', never the host 'time' buffer
```



Implemented in `StatementEvaluator._bind_series_name`: if the RHS has `.current` / `.history` / `.update` (a host or tracked series), the RHS is reduced to its **current scalar** before allocating or updating the LHS series. Same-bar reassignment (`x = 0` then `x := expr`) overwrites the current sample so `x[1]` remains the previous bar's final value. This prevents scripts such as dividend TTM time tracking from corrupting `time[j]` history when they rebind through a local.

UDF series parameters (0.6.2)

A Pine function parameter that is subscripted (`s[1]` inside `f(s) => ...`) is a **series of that argument**, not a scalar that happens to be named `s`. Interpret now allocates a per-call-site history wrapper when the caller passed a scalar or a host-series current (`src = close; updown(src)`). Compile already tracked UDF series state; the two hosts agree on official TradingView® builtins (P1p **0 MISMATCH**). Nested calls get distinct streams. See [Expressions & statements](#).

var / **varip** / **const**

Qualifier	Semantics in PYNE
<code>var</code> / <code>varip</code>	Initializer runs on first execution of that declaration site (tracked in <code>_var_declarations</code>), not only <code>bar_index == 0</code> . Later bars skip re-init so the value carries. History-tracked <code>var</code> series get a start-of-bar carry (<code>_commit_unwritten_history</code>) so <code>x[1]</code> is last bar's persist even when no <code>:=</code> ran this bar. On realtime ticks, <code>varip</code> re-evaluates the RHS.
<code>const</code> (v6)	Always initializes when the statement runs; not a cross-bar carry lock like <code>var</code> .
bare assign	Re-evaluates every bar; series "history" is the host series or prior list values.

Nested `var` inside `if barstate.islast` or a function therefore initializes on first path-taken bar—matching Pine's execution-based persistence.

na propagation

Operation	Rule
Binary arithmetic / comparison	Any <code>None</code> operand → <code>None</code> (or element-wise for lists)
Division by zero	→ <code>None</code> (not exception)
Soft type errors (<code>"a" + 1</code>)	→ <code>None</code>
Unary	<code>None</code> stays <code>None</code>
History OOB	→ <code>None</code>
Bare name <code>na</code>	Builtin / sentinel resolving to <code>None</code>
<code>nz(x, r)</code> (Numba path)	<code>nan</code> → replacement

List-valued series apply operators element-wise, aligning on the **trailing edge** when lengths differ (pad leading `None`).

`ta.highestbars` / `ta.lowestbars` offsets

These return a **bars-back offset** (not the extreme price):



Result	Meaning
0	Extreme is on the current bar
-1, -2, ...	Extreme is 1, 2, ... bars ago (down to <code>-(length-1)</code>)
-1 sentinel (warm-up / invalid / all-na)	Short history (<code>bar_index+1 < length</code>), bad length, or no finite samples in the window

On ties, the **oldest** extreme in the window wins (leftmost bar). Interpret and Numba kernels share this contract (`_highestbars` / `numba_highestbars`).

History subscript interaction: Pine history indices are non-negative (`series[n]` with $n \geq 0$). Negative indices soft-fail to `na` on interpret series wrappers so warm-up / auto-step-`-1` / `highestbars` misuse do not abort the bar loop. On the compile path, dynamic offsets are coerced float $\rightarrow$ int ($\text{NaN} \rightarrow 0$) and the computed array index is clamped to `[0, n]` so expressions like `high[-ta.highestbars(...)]` (future-looking form) soft-fail to `nan` near series end rather than raising.

Bar mode vs full-series mode

Technical helpers (`technical_submodules/core.py`) distinguish:

- **Full-series mode** (unit tests with explicit lists): `ta.sma` may return a full list of values.
- **Bar mode** (`_pine_bar_mode`): returns the **current scalar** so expressions like `ta.ema(close,12) - ta.ema(close,26)` stay numeric per bar.

History buffers for indicators are truncated to a rolling window (`_SERIES_MAX = 256` for wrapper histories) to avoid $O(n^2)$ full-history recomputation every bar.

Host series caps (`PYNE_SERIES_CAP`)

Pro API / `pynscript.runtime.Runtime` optionally **trims** chronological `current_series` lists and related host history so long charts do not grow ($O(\text{bars})$) memory per series.

Knob	Default	Meaning
<code>PYNE_SERIES_CAP</code>	on	Enable trimming. Disable with <code>0</code> / <code>false</code> / <code>no</code> / <code>off</code> (oracle / debug only).
Cap size	256 (<code>DEFAULT_SERIES_MAX</code>)	Raised when the script declares a larger <code>maxBarsBack=...</code> .
<code>PYNE_SERIES_MAX</code>	unset	Absolute override of the cap (positive int).
PineSeries history floor	1000	Separate from list caps; raised by <code>maxBarsBack</code> / <code>PYNE_SERIES_MAX</code> .

Correctness notes (see `src/pynscript/runtime/series.py` ; `backend/series.py` is a re-export shim):

- Window kernels (`ta.sma`, `highest`, ...) need `length ≤ cap`.
- Recursive smoothers under **incremental** TA (default `PYNE_TA_INCREMENTAL` on) carry state and stay safe independent of list length once warm.
- With **full recompute** (`PYNE_TA_INCREMENTAL=0`) and `bars ≫ cap`, EMA/RMA-style paths can diverge from a full-history oracle—prefer incremental (default) or raise the cap / disable series cap for goldens.

Out-of-range history offsets still return `na` (`None`); never `0`.



Unused derived series (0.3.10)

The interpret host updates `open` / `high` / `low` / `close` / `volume` / `time` every bar. Derived series (`h12`, `hlc3`, `ohl4`, `tr`, `time_close`) are written only when the source names them, uses `input.source`, or calls `ta.vwap` / `vwap` (default source is `hlc3` even when that identifier is absent). `ta.ao` rebuilds `h12` from high/low if the derived list is empty.

`PYNE_SERIES_RING` (chronological tail view) remains **default off**. When on, the host skips the second chronological list write.

Internals

Path	Role
<code>src/pynescrypt/ast/evaluator/names.py</code>	<code>visit_Name</code> , <code>visit_Attribute</code> , <code>visit_Subscript</code>
<code>src/pynescrypt/ast/evaluator/expressions.py</code>	NA-safe binary/unary ops, series coerce
<code>src/pynescrypt/ast/evaluator/statements.py</code>	<code>var</code> / <code>varip</code> / <code>const</code> assign
<code>src/pynescrypt/ast/evaluator/builtins/technical_submodules/core.py</code>	<code>_as_series</code> , bar mode finalize
<code>src/pynescrypt/ast/evaluator/builtins/utility.py</code>	<code>maxBarsBack</code> , <code>lastBarIndex</code> , <code>na</code> helpers
<code>src/pynescrypt/runtime/host.py</code>	Bar loop; derived-series skip; series-cap trim (<code>backend.runtime</code> re-exports)
<code>src/pynescrypt/runtime/series.py</code>	<code>PineSeries</code> , <code>resolve_series_cap</code> , <code>PYNE_SERIES_CAP</code> / <code>PYNE_SERIES_MAX</code> / <code>PYNE_SERIES_RING</code>

Compile path: series are `np.full(nBars, np.nan)` written at `__bar_idx`; history is `arr[__bar_idx - n]` with OOB → `np.nan` (see [Compiler overview](#)). Host `time_arr` is a chronological `float64` vector of bar-open Unix ms (same length as OHLCV); bare `time` lowers to `time_arr[__bar_idx]`, and `time[n]` uses the same history indexing rules as `close[n]`. When the host omits bar times, `Runtime` synthesizes `i * 60_000` so length always matches—calendar / `time` scripts stay defined on both modes.

Invariants & edge cases

- Chronology of `.history`**. Most-recent-first; reverse when converting to chronological lists for `ta.*` / `array.*`. Compile arrays are chronological (index 0 = first bar).
- No negative Pine indices**. Soft-fail to `na` / `nan` (not Python wraparound, and **not** a raise)—different from Python lists. See [highestbars](#) note above.
- None vs `float('nan')`**. Interpreter prefers `None`; Numba kernels use `np.nan`. Cross-mode comparisons must normalize.
- `maxBarsBack`**. Raises the host series cap when larger than the default 256; still not a TV-style hard chart depth, but it affects trim policy.



5. **PYNE_SERIES_CAP default on.** Disable only for debugging / full-history oracles; pair with incremental TA for production long histories.
6. **Derived skip.** `ta.vwap()` still requires a live `hlc3` series; the host treats `vwap` as a derived-series consumer.
7. **Tuple returns from multi-value `ta.*`.** Unpacking `[a, b, c] = ta.macd(...)` uses `current` when it is a sequence of matching arity; otherwise history heuristics apply.
8. **Host series assign is by value.** New names never share the host OHLCV/ `time` buffer reference.

Worked examples

History lag

```
//@version=6
indicator("lag")
delta = close - close[1]
plot(delta)
```

On bar 0, `close[1]` is `na` → `delta` is `na`. On bar 1+, arithmetic is ordinary floats (or series lists in batch tests).

var counter

```
//@version=6
indicator("count")
var int n = 0
n := n + 1
plot(n)
```

`n` initializes once; subsequent bars reassignment (`:=`) increments. Declaration sites are recorded in `_var_declarations`.

Element-wise NA

Given two list series of different lengths, `a + b` aligns tails and pads the longer prefix with `None`—preserving “most recent bars line up” intuition.



Failure modes

Symptom	Cause
All <code>na</code> after first bar	Host not updating series / <code>bar_index</code> between visits
<code>var</code> re-inits every bar	Fresh evaluator or <code>reset_var_declarations</code> each bar incorrectly
<code>TypeError</code> on <code>close + 1</code>	Series wrapper missing <code>.current</code> / not coerced
Compile vs interpret plot mismatch	<code>nan</code> vs <code>None</code> , or seed differences in EMA/RSI kernels — see parity
<code>time[j]</code> wrong after rebinding <code>last = time</code>	Expected only if host series were aliased; verify copy-on-assign path
<code>highestbars</code> always <code>-1</code>	Warm-up (<code>bar_index+1 < length</code>) or all- <code>na</code> window
Long-chart OOM / growing RSS	Series cap disabled or <code>PYNE_SERIES_MAX</code> huge; leave <code>PYNE_SERIES_CAP</code> on
EMA drift after many bars with cap	Full recompute + short cap — enable incremental TA or raise cap

See also

- [Expressions & statements](#)
- [Technical builtins](#)
- [Numba path](#)
- [Interpret ↔ compile parity](#)
- [Runtime hub](#)

EXPRESSIONS STATEMENTS

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title: "Expressions and statements" description: "How the AST walker evaluates operations, calls, control flow, assignments, and user definitions."

EXPRESSIONS AND STATEMENTS

Abstract

Evaluation is pure visitor dispatch: each ASDL node type maps to a `visit_*` method on a mixin. Expressions produce values (with Pine `na` and series rules); statements mutate context, register types/functions, or emit side effects via builtins. This page is the operational semantics of the interpreter path—what runs inside

`evaluator.visit(tree)` each bar.



Conceptual model

```
flowchart TB
    Script[Script body] --> Stmt[Statements]
    Stmt --> Assign[Assign / ReAssign]
    Stmt --> If[If / For / While / Once]
    Stmt --> Def[FunctionDef / TypeDef / EnumDef]
    Stmt --> ExprStmt[Expression statements]
    ExprStmt --> Expr[Expressions]
    Expr --> BinOp[BinOp / BoolOp / Unary / Compare]
    Expr --> Call[Call / Attribute / Name / Subscript]
    Call --> Builtin[Builtin map]
    Call --> UserFn[User function / method]
    Assign --> Ctx[(context dict)]
    UserFn --> Ctx
```

Interface surface

Expressions (`ExpressionEvaluator`)

Node	Behavior
<code>BoolOp</code> (<code>and</code> / <code>or</code>)	Manual short-circuit: <code>and</code> stops at first falsy (<code>na</code> / <code>0</code> / <code>False</code>); <code>or</code> stops at first truthy. Returns a Python <code>bool</code> .
<code>BinOp</code>	NA-safe <code>+</code> <code>-</code> <code>*</code> <code>/</code> <code>%</code> ; division by zero → <code>na</code>
<code>UnaryOp</code>	<code>not</code> , unary <code>+</code> <code>-</code> with NA and list map
<code>Compare</code>	Chained comparisons with NA-safe operators
<code>IfExp</code>	Ternary: condition, then, else
<code>Call</code>	Builtin dispatch, user functions, methods, multi-dispatch overloads
<code>Tuple</code> / lists	Literal sequences for unpacking and multi-arg returns

Binary ops coerce `PineSeries` -like objects to `.current` before operating. List operands broadcast or zip with trailing alignment (see [Series & history](#)).

Names (`NameEvaluator`)

Node	Behavior
<code>Name</code>	Context lookup; bare series builtins (<code>na</code> , <code>year</code> , ...); else string id (lazy)
<code>Attribute</code>	Qualified context keys, builtins (<code>strategy.position_size</code>), library modules, UDT fields/methods, array/drawing/matrix method markers, Python <code>getattr</code> fallback
<code>Subscript</code>	Series history / array / matrix indexing

Critical: qualified names like `strategy.opentrades.entry_price` are resolved via AST path construction (`ast_qualified_name`) **without** evaluating intermediate zero-arg series that would collapse the path to an int.



Statements (`StatementEvaluator`)

Construct	Behavior
<code>Script</code>	Walk body; finalize library exports if <code>library(...)</code> active
<code>Assign</code>	<code>var / varip</code> once-per-declaration; <code>const</code> ; ordinary assign; tuple unpack; <code>export</code> registration. History-tracked <code>var</code> series get a start-of-bar carry so <code>x[1]</code> is the prior persist.
<code>ReAssign (:=)</code>	Update existing binding (and UDT fields)
<code>If / For / While</code>	Standard control flow; loop variables in context
<code>Once</code>	Fire body the first time <code>test</code> is true (omit <code>test → true</code>) on a confirmed bar, then skip. Unconfirmed realtime ticks may re-run until <code>barstate.isconfirmed</code> commits the fired flag. Does not return a value.
<code>FunctionDef</code>	Store callable in context; multi-dispatch overload lists; <code>export</code> ; methods tagged <code>__pine_method__</code> . After bar 0 the host walks a hot body that skips Function/Type/Enum/Import (already bound).
<code>TypeDef / EnumDef</code>	Type registry / enum member dicts
<code>Import</code>	Resolve <code>LibraryRegistry</code> → bind alias to <code>LibraryModule</code>
Declarations	<code>indicator / strategy / library</code> via declaration builtins

Function invocation binds parameters into context (with restore of prior bindings on exit) so nested calls and bar-local state interact cleanly. After bar 0, hosts set `_pine_defs_locked` so re-visiting the script does not re-register overload tables. The locked walk inlines `Assign` and `Expr(Call)` (no generic `visit` frame) and UDF/call-expression history keys use `_pine_site_id` stamped on the AST.

Calls and multi-dispatch

Method markers returned from attribute evaluation:

Marker	Meaning
<code>("_method_call", instance, name)</code>	UDT method
<code>("_array_method", list, name)</code>	<code>array.<name>(self, ...)</code>
<code>("_ns_method", obj, "label.get_text")</code>	Drawing/matrix namespace method
<code>("_ext_method", receiver, name)</code>	Free <code>method</code> with receiver as first arg

Overloads with typed first parameters (e.g. `matrix.float` vs `array.label`) select via coarse type tags; `na` receivers deliberately exclude matrix/array/drawing tags so optional UDT fields do not mis-dispatch to `tostring` on collections.



Internals

Path	Role
src/pynescrypt/ast/evaluator/expressions.py	Ops, calls, conditionals
src/pynescrypt/ast/evaluator/names.py	Names, attributes, subscripts
src/pynescrypt/ast/evaluator/statements.py	Script structure, assign, defs, import
src/pynescrypt/ast/evaluator/literals.py	Constants, colors, strings
src/pynescrypt/ast/evaluator/base.py	Context, math constants, library registry hook
src/pynescrypt/ast/evaluator/types.py	EvaluatorProtocol typing for mixins

Invariants & edge cases

1. **Context is the single store.** Builtins read OHLCV and strategy series from `self.context` ; hosts must keep it coherent each bar.
2. **String fallback names.** An unbound `Name` returns its id string so later attribute/call resolution can still hit the builtin map (`"ta.sma"` patterns via attribute chains).
3. **Parameter unbind.** Missing-before-bind params use a sentinel so unbind `pop` s rather than leaving `None` ghosts.
4. **na normalization.** String `"na"` / `"nan"` / `"none"` may coerce to `None` at dispatch boundaries for UDT optional args.
5. **User functions re-enter the visitor.** Recursive and higher-order patterns are limited by Python stack, not a Pine-specific trampoline.

Worked examples

Ternary and comparison chain

```
//@version=6
indicator("cmp")
v = close > open ? close - open : open - close
plot(v)
```

Each operator path is NA-safe: if `close` or `open` were missing, comparisons and arithmetic yield `na` .

User function with series

```
//@version=6
indicator("fn")
f(x, n) =>
    ta.sma(x, n)
plot(f(close, 14))
```

`f` is stored as a Python callable that rebinds `x` / `n` and visits the function body AST each call.



UDF series parameters (0.6.2): when a caller passes a scalar (or a host-series current) into a function that then reads `s[1]` , interpret wraps that parameter as a per-call-site history stream. `src = close; updown(src)` therefore matches compile — the previous-bar sample is the previous argument at that call site, not an empty scalar. Official TradingView® builtin scripts are 0 MISMATCH vs compile on this contract.

once (August 2026)

```
//@version=6
indicator("once")
var int n = 0
once close > ta.sma(close, 20)
    n := 1
plot(n)
```

Equivalent to a `var bool` latch plus `if` , but cheaper: the structure is not re-evaluated after it fires on a closed bar. On a forming realtime bar (`Runtime.run(..., realtime_last_bar=True, realtime_ticks=N)`), the latch is not committed until `barstate.isconfirmed` . Compile is historical (`isconfirmed` always true).

Intrabar rollback on realtime ticks (0.6.0)

On a forming realtime bar, intermediate ticks snapshot the `var` scope (context bindings, series currents, declaration set) and restore it after the visit — every tick starts from confirmed state and only the final tick commits. `varip` names persist across ticks by design. Shallow by design: history pushes, the series-assign map, in-place container mutations, and broker side effects are not rolled back.

Method multi-dispatch sketch

```
method log(this, string s) => ...
method log(this, float x) => ...
```

Both register under the same name with `__pine_overloads__` ; call sites pick by receiver/arg type tags.

Failure modes

Error	Meaning
<code>unexpected type of node</code>	Missing <code>visit_*</code> for an AST construct
<code>Unsupported binary operator</code>	Op class not in the dispatch table
History <code>series[-1]</code> is <code>na</code>	Negative Pine offsets soft-fail to <code>None</code> (not Python wraparound, not a raise)
Wrong overload / destroyed receiver	Historical <code>na</code> string path; now normalized—report residual mismatches as bugs
Stack overflow	Deep recursion in user functions

See also

- [Series & history](#)
- [Builtins hub](#)
- [Libraries](#)
- [Type system \(core\)](#)

OVERVIEW

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title: "Builtins" description: "Dispatch architecture for ta., *strategy*., collections, plotting, request.*, and other standard namespaces."

BUILTINS

Abstract

Pine's standard library is not a single module: it is a **flat qualified-name dispatch table** (`"ta.sma"` , `"strategy.entry"` , `"array.push"` , ...) assembled from category mixins. Each entry is a handler `(args, kwargs?)` → `value` with side effects when the language requires them (orders, plots, drawings, inputs). This hub orients the surface; child pages go deep on technical, strategy, collections, drawing/plotting, and request/input.

Conceptual model

```

flowchart LR
    Call[Call / Attribute] --> Qual[Qualified name]
    Qual --> Map["_build_builtin_map()"]
    Map --> Handler[BuiltinHandler]
    Handler --> Val[Return value]
    Handler --> Side[Side effects]
    Side --> Strat[StrategyState]
    Side --> Plot[PlotRegistry]
    Side --> Draw[DrawingRegistry]
    Side --> Inp[_input_declarations]

```

`BuiltinEvaluator` merges mixin maps in a fixed order and then registers ticker, logging, color, timeframe, and script-declaration functions. The `strategy()` declaration handler is wrapped so broker settings apply to `StrategyState`.

Interface surface

Namespace / area	Mixin / module	Docs
<code>ta.*</code>	<code>TechnicalAnalysisMixin</code> + <code>technical_submodules/*</code>	Technical
<code>strategy.*</code>	<code>StrategyBuiltinsMixin</code> , <code>StrategyConstantsMixin</code>	Strategy
<code>array.*</code> / <code>matrix.*</code> / <code>map.*</code>	<code>ArrayBuiltinsMixin</code> , matrix/map evaluators	Collections
<code>plot*</code> / <code>hline</code> / <code>line</code> / <code>label</code> / ...	<code>PlottingFunctionsMixin</code> , <code>DrawingBuiltinsMixin</code>	Drawing & plotting
<code>request.*</code> / <code>input.*</code>	<code>RequestBuiltinsMixin</code> , <code>InputBuiltinsMixin</code>	Request & input
<code>math.*</code> , min/max, <code>nz</code> , ...	<code>NumericBuiltinsMixin</code>	(this page)
<code>str.*</code>	<code>StringBuiltinsMixin</code>	(this page)
<code>color.*</code>	<code>register_color_functions</code>	(this page)
<code>syminfo</code> / <code>ticker.*</code>	<code>register_ticker_functions</code>	(this page)
<code>timeframe.*</code>	<code>register_timeframe_functions</code>	(this page)
<code>alert</code> / logging	<code>AlertsMixin</code> , <code>register_logging_functions</code>	Alerts
<code>indicator</code> / <code>strategy</code> / <code>library</code>	<code>register_script_declaration_functions</code>	Libraries
Footprint / volume_row	<code>FootprintBuiltinsMixin</code>	Request & input

Numeric and string essentials

- **Numeric:** `math.*` constants from `BaseEvaluator` (`math.pi`, `math.e`, golden-ratio helpers); functions cover abs, min/max, rounding, logs, trig, and NA helpers (`nz`, `na` predicates via utility).
- **String:** `str.*` formatting, conversion (`str.toString` respects `format.*` constants), and manipulation aligned with common Pine scripts in the builtin corpus.

Color, ticker, timeframe

- **Colors:** hex constants (`color.red` → `#F23645`, etc.) plus composition helpers.



- **syminfo / ticker:** host-provided `Syminfo` object or flat context keys; ticker construction for `request.security` symbols.
- **timeframe:** period flags (`isintraday`, `isdaily`, ...) default daily in base constants; hosts override from bar spacing.

Alerts

`alert(message, freq)` and `alertcondition(condition, title, message)` record structured firings (`message`, `freq`, `bar_index`, `time`, `source`) with TV-style frequency rules (`once_per_bar`, `once_per_bar_close`, `all`). Hosts export them on `/run` and can POST last-bar batches to webhooks (**L2**). Side channel only — **not** strategy events.

Full detail: [Alerts](#).

Internals

Path	Role
<code>src/pynescrypt/ast/evaluator/builtins/__init__.py</code>	<code>BuiltinEvaluator</code> composition
<code>src/pynescrypt/ast/evaluator/builtins/base.py</code>	Dispatch mixin, <code>_call_builtin</code> , registration
<code>src/pynescrypt/ast/evaluator/builtins/*.py</code>	Category implementations
<code>scripts/generate_builtin_metadata.py</code>	LSP/metadata surface (not hand-edited JSON)

Handlers are looked up by the fully qualified string built during name/attribute evaluation. Zero-arg series (e.g. `strategy.equity`) are registered as builtins that ignore empty args.

Invariants & edge cases

1. **One map, many mixins.** New builtins require a map entry *and* (usually) metadata regeneration for LSP.
2. **Kwargs and positionals.** Handlers accept both; Pine named arguments arrive as kwargs when the call AST carries them.
3. **Bar mode.** Stateful `ta.crossover` / similar use per-bar call indices reset by the host (`_cross_call_i`). Incremental `ta.*` (`PYNE_TA_INCREMENTAL`, default on) uses a sibling `_ta_call_i` slot.
4. **Mock fallbacks.** Bare equity-style `request.security` strings may still mock offline; foreign / fundamental prefixes and complex expressions return `na` (see [request / input](#)).
5. **Compile path subset.** Numba builtins reimplement only a fraction of this table; object mode covers more via Python helpers—see [Compiler](#).

Worked example — resolution path

```
plot(ta.sma(close, 14))
```

1. Attribute/call chain builds qualified name `ta.sma`.
2. `_call_builtin("ta.sma", [close_series, 14])`.
3. Handler coerces series → list, computes SMA, finalizes scalar or list.
4. `plot` registers a `Plot` with that series value and returns the plot id.



Failure modes

Symptom	Fix
Unknown built-in function	Typo, version surface gap, or failed attribute chain
Always mock prices	Wire <code>data_feed</code> / <code>data_provider</code> into evaluator context
LSP knows a function runtime lacks	Metadata ahead of implementation—check missing features
Handler arity errors	Positional vs keyword mismatch in script

See also

- [Technical](#)
- [Strategy](#)
- [Collections](#)
- [Drawing & plotting](#)
- [Request & input](#)
- [Alerts](#)
- [Builtin metadata \(LSP\)](#)

TECHNICAL

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title: "Technical analysis (ta.)" *description: "ta. indicators: series helpers, bar mode, interpret ↔ compile parity, and submodule layout."*

TECHNICAL ANALYSIS (`ta.*`)

Abstract

The `ta.*` namespace is the largest pure-compute surface in PYNE: moving averages, oscillators, volatility bands, volume studies, pattern helpers, and cross/rise/fall predicates. Implementations live under `technical_submodules/` and are composed into `TechnicalAnalysisMixin`. Numerical behavior is validated against TradingView reference series (see [numerical validation](#)); the design goal is IEEE-limit parity, not “approximate TA.” Interpret and compile (Numba) hosts share the same formulas for the kernels listed under [Interpret ↔ compile parity](#).



Conceptual model

```
flowchart TB
    Args[Args: series, period, ...] --> As[_as_series]
    As --> Chrono[Chronological list]
    Chrono --> Inc{bar mode + incremental?}
    Inc -->|yes hot path| State[Call-site state 01]
    Inc -->|no| Kernel[Full-history kernel]
    State --> Scalar[Current scalar]
    Kernel --> Fin[_finalize_series]
    Fin -->|bar mode| Scalar
    Fin -->|test mode| Full[Full list]
```

Hosts in production set bar mode so indicator calls return **scalars for the active bar**, matching how Pine expressions compose (`ta.ema(a) - ta.ema(b)`).

Incremental hot path

When `_pine_bar_mode` and `_pine_ta_incremental` are enabled (Runtime default; disable with `PYNE_TA_INCREMENTAL=0`), hot `ta.*` builtins update **call-site state** once per bar instead of recomputing full history. Call sites are indexed like crossovers (`_ta_call_i` reset each bar). Nested forms such as `ta.ema(ta.sma(close, 14), 10)` stay correct because each call keeps its own slot. Golden suite: `tests/test_ta_incremental.py` (incremental last values $\equiv$ full recompute).

Family	Incremental builtins (non-exhaustive)
Moving averages	<code>sma</code> , <code>ema</code> , <code>rma</code> , <code>wma</code> , <code>hma</code> , <code>vwma</code> , <code>swma</code> , <code>kama</code> , <code>dema</code> , <code>tema</code> , <code>alma</code>
Oscillators	<code>rsi</code> , <code>macd</code> , <code>stoch</code> / <code>stochrsi</code> , <code>cci</code> , <code>cmo</code> , <code>tsi</code> , <code>roc</code> , <code>wpr</code> , <code>aroon</code> , <code>dpo</code> , <code>kst</code>
Volatility / structure	<code>atr</code> , <code>tr</code> , <code>stdev</code> , <code>bb</code> , <code>highest</code> / <code>lowest</code> , <code>linreg</code> , <code>adx</code> / <code>dmi</code> , <code>supertrend</code> , <code>kc</code> , <code>sar</code> , <code>donchian</code>
Volume (0.3.10–0.3.11)	<code>obv</code> , <code>wad</code> / <code>wvad</code> , <code>cmf</code> , <code>klinger</code> , <code>mfi</code> , <code>vwap</code> , <code>accdist</code> , <code>nvi</code> , <code>pvi</code>
Other	<code>change</code> / <code>mom</code> / <code>cum</code> , <code>median</code> , <code>percentrank</code> , <code>rising</code> / <code>falling</code> , <code>barssince</code> , <code>pivots</code>

`ta.nvi` / `ta.pvi` are incremental ($O(1)$ /bar; `PYNE_TA_INCREMENTAL=0` full-list). **`ta.atr` is Wilder RMA of true range** on interpret (full + incremental) and Numba (`ta.rma(ta.tr, length)` — 0.3.4+). Supertrend is locked $\text{mid} \pm \text{factor} \cdot \text{ATR}$ (not the TV band ratchet).

Interface surface

Dispatch keys (non-exhaustive; map is authoritative in `technical.py`):



Family	Examples
Moving averages	<code>ta.sma</code> , <code>ta.ema</code> , <code>ta.wma</code> , <code>ta.rma</code> , <code>ta.hma</code> , <code>ta.vwma</code> , <code>ta.swma</code> , <code>ta.alma</code>
Oscillators	<code>ta.rsi</code> , <code>ta.stoch</code> , <code>ta.cci</code> , <code>ta.cmo</code> , <code>ta.mfi</code> , <code>ta.roc</code> , <code>ta.wpr</code> , <code>ta.tsi</code> , <code>ta.rci</code>
Trend / channels	<code>ta.macd</code> , <code>ta.adx</code> , <code>ta.dmi</code> , <code>ta.supertrend</code> , <code>ta.sar</code> , <code>ta.linreg</code>
Volatility	<code>ta.atr</code> , <code>ta.tr</code> , <code>ta.bb</code> , <code>ta.bbw</code> , <code>ta.kc</code> , <code>ta.kcw</code> , <code>ta.stdev</code> , <code>ta.variance</code>
Volume	<code>ta.obv</code> , <code>ta.vwap</code> , <code>ta.mfi</code> (shared), volume submodule helpers
Structure	<code>ta.highest</code> / <code>lowest</code> / <code>highestbars</code> / <code>lowestbars</code> , <code>ta.pivohigh</code> / <code>pivotlow</code>
Predicates	<code>ta.crossover</code> , <code>ta.crossunder</code> , <code>ta.cross</code> , <code>ta.rising</code> , <code>ta.falling</code>
Stats / other	<code>ta.change</code> , <code>ta.mom</code> , <code>ta.cum</code> , <code>ta.median</code> , <code>ta.mode</code> , percentiles, <code>ta.barssince</code> , <code>ta.valuewhen</code> , <code>ta.correlation</code>

Argument conventions

- Typical form: `ta.fn(series, length)` .
- Some functions allow **period-only** calls (e.g. `ta.highest(20)`) defaulting source to `high` / context series via `_expect_series(..., allow_period_only=True)` .
- Multi-value returns (e.g. MACD, BB, Supertrend) unpack as tuples/lists; assignment uses StatementEvaluator unpack rules.
- Fractional lengths floor to int (TradingView-compatible).

Stateful crosses

`ta.crossover` / `ta.crossunder` need previous-bar pairs. In bar-mode runs the host resets a call-site index (`_cross_call_i`) each bar so multiple cross calls in one script keep independent state.



Internals

Path	Role
<code>src/pynescrypt/ast/evaluator/builtins/technical.py</code>	Dispatch map aggregation
<code>.../technical_submodules/core.py</code>	Series coerce, expect helpers, bar finalize, incremental kernels
<code>.../moving_averages.py</code> , <code>oscillators.py</code> , <code>volatility.py</code> , <code>volume.py</code> , ...	Kernels + inc wiring
<code>.../common.py</code> , <code>basic.py</code> , <code>advanced.py</code> , <code>patterns.py</code> , <code>strategies.py</code> , <code>synthesizer.py</code> , <code>economics.py</code>	Additional families
<code>src/pynescrypt/compiler/numba_builtins.py</code>	Compile-path Numba kernels (must track interpret formulas)
<code>tests/test_ta_indicators_*.py</code> , <code>tests/test_indicators.py</code>	Regression
<code>tests/test_ta_incremental.py</code>	Inc $\equiv$ full recompute golden (full CI gate)
<code>tests/test_first_party_ta_goldens.py</code>	Dual-host first-party goldens (ATR / Supertrend / Keltner)
<code>tests/test_interp_compile_parity.py</code>	Always-on dual-host smoke + full-corpus mark
<code>scripts/compare_interp_compile.py</code>	Corpus harness (report under <code>.cache/interp_compile_parity.json</code>)
<code>tests/test_compiler_numba.py</code> (<code>TestInterpCompilePlotParityFixes</code> , highestbars offsets, ...)	Targeted formula locks
<code>docs/numerical_validation_report.md</code>	Published precision summary (TV / IEEE bounds)

History for wrapper series is reversed to chronological order and may be truncated (`_SERIES_MAX`) before full kernels run. Incremental path only needs `series[-1]` per call, so truncation does not freeze state.

Interpret $\leftrightarrow$ compile parity

Numba kernels are **separate code** from the interpreter, but the dual-host contract is **same formula, same warm-up mask, same na rules** for the surface below. Drift is a bug, not an allowed “approx compile.” Prove regressions with the [interp \$\leftrightarrow\$ compile harness](#) and the [numerical validation](#) page (TV / IEEE bounds).



Topic	Shared behavior
<code>ta.rsi</code>	Wilder on both hosts: SMA seed of the first <code>period</code> deltas, then RMA of gains/losses. First finite bar is index <code>period</code> (<code>period</code> deltas need <code>period+1</code> prices). Not a simple window average.
<code>ta.roc</code>	Standard TV formula <code>100 * (src - src[length]) / src[length]</code> . Warm-up is <code>na</code> , never an early <code>0.0</code> . Zero or missing baseline → <code>na</code> . Interpret no longer used a wrong lookback denominator or early zero.
<code>ta.wma</code>	Requires a full non-<code>na</code> window of length <code>N</code> . Nested forms such as <code>ta.wma(ta.roc(...), ...)</code> stay <code>na</code> until the inner series has produced <code>N</code> finite samples (no partial-window reweight).
<code>ta.cum</code>	Running sum; Pine <code>na</code> / IEEE NaN treated as 0 (skipped contribution), matching TradingView cumulative-sum semantics.
<code>ta.highestbars</code> / <code>ta.lowestbars</code>	Return negative bars-back offsets : <code>0</code> if the extreme is the current bar, <code>-1</code> one bar ago, ..., <code>-(length-1)</code> at the far edge. Short history (<code>i + 1 < length</code>) returns <code>-1</code> . Matches Aroon-style scripts that index with <code>high[ta.highestbars(...)] / math.abs(ta.lowestbars(...))</code> .
<code>math.avg</code> (multi-arg)	Arithmetic mean of the arguments. Any <code>na</code> argument → <code>na</code> (do not skip). Not a rolling <code>ta.sma</code> .
<code>ta.linreg</code>	Least-squares fit over the window; endpoint at <code>x = n-1</code> when <code>offset=0</code> (TV): <code>mean_y + slope * ((n-1) - mean_x)</code> / compile <code>intercept + slope * (n - 1 - offset)</code> . <code>Length < 2</code> → <code>na</code> .
<code>ta.mfi</code>	Warm-up aligned on both hosts: needs <code>length + 1</code> typical-price samples (direction vs previous bar). Equal typical prices contribute to neither side; only-pos / only-neg MF → <code>100 / 0</code> (or <code>50</code> when both empty).
<code>ta.rci</code>	Rank Correlation Index (Spearman of time vs value ranks) is implemented on the compile path (<code>numba_rci</code>) as well as interpret; stays in numeric mode when called.

0.3.4+: `ta.atr` is Wilder RMA of TR on interpret + Numba (not EMA-of-TR). EMA dual-host seed is full-list SMA seed matching incremental/Numba. First-party dual-host goldens (ATR / Supertrend / Keltner) and `test_ta_incremental` gate residual drift. Any remaining gaps are tracked under **missing features**—not as silent “close enough” for the rows above.

Parity harness

```
# Always-on smoke (stable scripts under tests/test_interp_compile_parity.py)
pytest tests/test_interp_compile_parity.py -q

# Corpus compare (default ~50 scripts × 1000 bars; writes report JSON)
python scripts/compare_interp_compile.py --bars 1000 --limit 50
python scripts/compare_interp_compile.py --glob 'average_*.pine' --bars 200

# Optional longer pytest path
pytest tests/test_interp_compile_parity.py -m interp_compile_full

# Formula locks (RSI/ROC/WMA/cum/math.avg/highestbars, ...)
pytest tests/test_compiler_numba.py -k 'InterpCompilePlotParity or highestbars_lowestbars_negative'
```

Report path: `.cache/interp_compile_parity.json`. Methodology and acceptable relative-error bands vs TradingView: **Numerical validation**.



Invariants & edge cases

1. **Warm-up** → `na`. Insufficient bars (e.g. SMA length (N) needs (N) samples; RSI needs `period+1` prices) yield `None` / leading `na` in full-series mode.
2. **NA in window**. Many kernels propagate `na` if any window element is missing (SMA-like sums, **WMA**, MFI money-flow samples). **ta.cum is the TV exception** (`na` → 0 contribution).
3. **Default sources**. `ta.atr` pulls high/low/close from `current_series` / context when not passed explicitly.
4. **Compile kernels must track interpret**. Numba `numba_*` implementations are separate source; for the parity table above they are formula-locked by dual-host tests—do not “approximate” for speed.
5. **No chart look-ahead**. Kernels only see history available at the current bar index; `request.security` lookahead is a separate concern.
6. **Incremental** = **full recompute (oracle)**. ATR seed is already Wilder RMA on both hosts; changing seed rules is a correctness project, not a silent perf flag.

Worked examples

Classic overlay

```
//@version=6
indicator("SMA 14")
v = ta.sma(close, 14)
plot(v)
```

In bar mode each visit returns one float (or `na`); the host stacks them into a series for the response envelope.

Cross entry signal

```
//@version=6
strategy("X")
fast = ta.ema(close, 12)
slow = ta.ema(close, 26)
if ta.crossover(fast, slow)
    strategy.entry("L", strategy.long)
```

Cross state is bar-local; ensure the runtime resets cross call indices when reusing an evaluator.

Multi-value unpack

```
[macdLine, signal, hist] = ta.macd(close, 12, 26, 9)
plot(hist)
```



Failure modes

Symptom	Cause
Always <code>na</code>	Length longer than available history; or series not updated
Wrong vs TV by large margin	Wrong source series; mock OHLCV; or non-bar-mode list composition bug
Cross never fires	Call-index state not reset / shared incorrectly across bars
Compile diverges from interpret	Kernel drift in <code>numba_builtins.py</code> (RSI must stay Wilder; ROC warm-up must stay <code>na</code> ; WMA full window; highestbars sign; <code>math.avg</code> <code>na</code> rule)—re-run the parity harness
Early ROC / nested WMA finite too soon	Legacy quirks: ROC returned <code>0.0</code> before lookback, or WMA reweighted partial/ <code>na</code> windows—fixed on both hosts
<code>high[ta.highestbars(...)]</code> mostly <code>na</code>	Offsets are negative ; only offset <code>0</code> indexes the current bar without <code>math.abs</code>
<code>math.avg</code> finite when an arg is <code>na</code>	Bug: multi-arg avg must propagate <code>na</code> , not skip

See also

- [Series & history](#)
- [Strategy builtins](#)
- [Numba path](#)
- [Numerical validation](#)
- [Missing features](#) (residual dual-host gaps)
- [Compiler overview](#)

STRATEGY

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title: "Strategy builtins" description: "Orders, fills, OCA, commission, risk gates, position metrics, and StrategyState."

STRATEGY BUILTINS

Abstract

PYNE's strategy layer is a **per-run broker simulation** driven by `strategy.*` calls during the bar loop. It is not a live exchange adapter: orders become fills against bar OHLC (market immediately; limit/stop via pending book), commissions and slippage adjust PnL, and risk helpers can block entries. Structured `StrategyEvent` records form the parity contract with `pyne-worker` and the HOOX trade mesh.



Conceptual model

```
sequenceDiagram
    participant Host as Bar loop
    participant Pend as Pending orders
    participant Script as strategy.* body
    participant State as StrategyState
    participant Ev as Event buffer
    Host->>Pend: process_pending_orders(OHLC)
    Pend->>State: fills / OCA
    Pend->>Ev: order/entry/cancel...
    Host->>Script: visit(AST)
    Script->>State: entry/exit/close/order/risk
    Script->>Ev: StrategyEvent
    Host->>Ev: drain_events()
```

Fill-before-script matches the interpreter `Runtime` and the compile-path `strategy broker`.

Interface surface

Order placement

Builtin	Role
<code>strategy.entry</code>	Open/add (or reverse) by direction; market or pending limit/stop
<code>strategy.exit</code>	Bracket-style exit (pending limit/stop + OCA); market exit when no levels
<code>strategy.close</code> / <code>strategy.close_all</code>	Flatten by id or all (<code>qty_percent</code> supported)
<code>strategy.order</code>	Lower-level order with OCA name/type, partial fill cap
<code>strategy.cancel</code> / <code>strategy.cancel_all</code>	Remove pending orders

Directions accept Pine constants (`strategy.long` / `strategy.short`) and common string aliases.

`strategy.exit` surface (0.3.4+)

Feature	Status
Pending stop/limit brackets + OCA	Interpret + compile — fills via <code>process_pending_orders</code> OHLC path
<code>from_entry</code> multi-leg filter	Interpret + compile — unknown id is soft no-op
<code>qty_percent</code>	Interpret + compile <code>close</code> — % of target (whole pos or <code>from_entry</code>); wins over <code>qty</code> ; ≤ 0 /na → no-op; >100 capped
Trail (<code>trail_offset</code> / <code>trail_points</code> / optional <code>trail_price</code>)	Interpret + compile — distances are ticks ($\times$ <code>mintick</code>). Pending stop ratchets from bar high/low (OHLC approx). <code>trail_points=0</code> / <code>na</code> is ignored so a valid <code>trail_offset</code> still applies (0.3.8).
<code>profit</code> / <code>loss</code>	Ticks from the exit's entry average ($\text{ticks} \times \text{mintick}$). Long: <code>entry ± offset</code> ; short flips the sign. <code>na</code> / ≤ 0 ignore that leg. Absolute <code>limit</code> / <code>stop</code> win when both are set.



```
//@version=6
strategy("Exit demo", overlay=true, initial_capital=100000)
if bar_index == 5
    strategy.entry("L", strategy.long, qty=10)
if bar_index == 6
    strategy.exit("X", from_entry="L", stop=low * 0.98, limit=high * 1.02, qty_percent=50)
// Tick form (same surface): profit/loss are offsets from entry avg, not prices.
// strategy.exit("X", from_entry="L", profit=200, loss=100)
```

Position and performance series

Zero-arg builtins include `strategy.position_size` (signed: +long / -short), `position_avg_price`, `position_entry_name`, `opentrades` / `closedtrades` counts, `netprofit`, `openprofit`, `equity`, `cash`, gross win/loss stats, averages, max drawdown/runup, and max contracts held.

`strategy.position_avg_price` is the VWAP of **open** entry fills. A filled position is **not** overwritten when `strategy.entry` is called again with the same id (`if cond: strategy.entry("L", strategy.long)` stays at the first fill when `pyramiding=0`). Extra same-direction fills only occur when `opentrades < pyramiding + 1` (PYNE cap = `pyramiding + 1`; TV `pyramiding` default 1 $\equiv$ PYNE 0). Pending limit/stop orders with that id are still replaced. When flat the series is `na`.

Trade queries

Indexed accessors:

- `strategy.closedtrades.entry_*` / `exit_*` / `profit` / `size` / `commission`
- `strategy.opentrades.entry_*` / `size` / `profit` / `commission`
- Open/closed trade fields include bar/time/id/comment plus approximate MAE/MFE (`max_drawdown` / `max_runup` from bar high/low)

Risk (interpret + compile halt cascade)

Builtin	Effect
<code>strategy.risk.max_position_size</code>	Cap size as % of equity
<code>strategy.risk.max_intraday_loss</code>	Intraday loss gate — percent-of-initial-capital default, <code>strategy.cash</code> absolute (0.6,0, both brokers)
<code>strategy.risk.max_intraday_filled_orders</code>	Order count cap (day-scoped, enforced on both brokers)
<code>strategy.risk.max_drawdown</code>	Absolute / percent drawdown halt
<code>strategy.risk.max_cons_loss_days</code>	Consecutive losing calendar days → <code>entries_blocked</code>
<code>strategy.risk.allow_entry_in</code>	"all" "long" "short"

Blocked entries still emit a diagnostic-style event with comment `risk_blocked` where implemented. Compile broker shares the common halt cascade above (not a full TradingView® risk engine).

Declaration

`strategy(title, ...)` applies broker settings onto `StrategyState`: `initial_capital`, commission type/value, slippage ticks, pyramiding, `avg_price_model`, etc.



Average price model (pynescrypt extension)

Not part of official TradingView Pine. Controls how `strategy.position_avg_price` evolves when size changes:

avg_price_model	On same-direction adds	On partial reduce
"stock" (default)	Arithmetic VWAP of fills	Re-average remaining open legs (FIFO multi-trade style)
"futures"	Same arithmetic VWAP	Sticky net AEP until flat (linear USDT-M / BTCUSDT-like)
"inverse"	Arithmetic today; harmonic add planned	Sticky until flat (same reduce rule as futures)

```
strategy("Perp style", avg_price_model="futures")
// or tokens:
// strategy(..., avg_price_model=strategy.avg_price_futures)
```

Linear add formula (stock and futures):

$$[\text{avg}' = \frac{\text{avg} \cdot |\text{size}| + p \cdot q}{|\text{size}| + q}]$$

Under "futures", partial closes realize PnL against the sticky average (not per-leg FIFO entry prices), and closed-trade `entry_price` records that same average. Commission is never folded into the average numerator.

There is no `strategy.order.avg_future_price` series—use `strategy.position_avg_price` with the mode switch.

Leverage (pynescrypt extension — simpler futures UI)

Exchanges expose **leverage** (e.g. 5× / 10× / 20×). Prefer that over TV-style `margin_long` / `margin_short` percentages:

```
//@version=6
// TV-safe: strategy() kwargs need *const* (literals / const vars).
// Host UI can still inject runtime input overrides into PYNE.
strategy(
    "BTCUSDT style",
    avg_price_model="futures",
    leverage=10,
    default_qty_type=strategy.percent_of_equity,
    default_qty_value=100, // use full equity as *margin*
    initial_capital=10_000)

// Optional Inputs-tab control for *trade logic* (not TV strategy Properties):
// lev = input.float(10, "Leverage", minval=1, maxval=125)
```

Setting	Effect
<code>leverage=N</code> (default 1)	Buying power multiplier
<code>percent_of_equity</code> / <code>cash</code> default qty	$\text{qty} = \text{margin} \times \text{leverage} / \text{price}$
<code>fixed</code> default qty	Unchanged (contracts as written)
<code>strategy.cash</code> / capital held	Margin locked = notional / leverage
<code>strategy.margin_liquidation_price</code>	Simple isolated estimate when <code>leverage > 1</code>
<code>strategy.leverage</code>	Read-back of configured multiplier



`leverage=10` also sets internal margin % to $100/10 = 10$ (TV `margin_long / margin_short` equivalent). If only `margin_long` is set, leverage is derived as $100 / \text{margin_long}$.

input before strategy() — TV vs PYNE

	TradingView	PYNE
Statements before <code>strategy()</code>	Allowed for const vars (and types/enums/imports)	Allowed
<code>input.*()</code> return qualifier	input (not const)	Plain Python value (no qualifier system)
<code>strategy(leverage=input.float(...))</code>	Compile error — most <code>strategy()</code> params require const	Interpret: works. Compile: folds const-like defval into the broker ctor (<code>input.float(10)</code> → <code>leverage=10</code>)
User-adjustable leverage on TV	Settings → Properties (“Long/Short leverage”, mapped from <code>margin_long / margin_short</code>)	Declaration <code>leverage=</code> or host-injected inputs

Qualifier hierarchy on TV: `const < input < simple < series`. A parameter that needs **const** **cannot** accept an `input` value. So even if `lev = input.float(10)` appears *before* `strategy()`, you still cannot pass `lev` into `strategy(...)` on TV.

PYNE does **not** enforce qualifiers; for compile-path parity with TV’s `const` rule, non-const strategy kwargs are skipped unless their series folds to a constant defval.

Internals

StrategyState (strategy.py)

Per-evaluator instance fields include:

- Position: direction (`flat / long / short`), size (non-negative internal), entry metadata
- Books: `pending_orders`, `open_trades`, `closed_trades`
- Economics: capital, commission model, slippage, mintick
- Risk: max position %, drawdown, consecutive loss days, `entries_blocked`
- Equity curve peak/trough for max DD / runup
- `_events: list[StrategyEvent]` drained each bar

`signed_position_size()` implements Pine’s signed `strategy.position_size`.

Order and fills

```
Order: market | limit | stop | stop-limit
+ oca_name / oca_type ∈ {none, cancel, reduce}
+ max_fill_per_bar (0 = fill remaining)
```

`process_pending_orders(open, high, low, close)` walks the book, computes trigger prices from OHLC (gap-aware open logic for limits/stops), applies commission/slippage, updates trades, runs OCA side effects, and emits events.



OCA

oca_type	On fill of a group member
cancel	Cancel other pending in group
reduce	Reduce sibling quantities by fill size
none	Independent

Covered by `tests/test_oca_commission.py` and order-fill suites.

Invariants & edge cases

1. **Isolation.** Never use class-level strategy state; concurrent runs must not share books.
2. **Pyramiding.** Additional entries respect `pyramiding` from the declaration.
3. **Partial fills.** `max_fill_per_bar` / defaults allow multi-bar completion of large orders.
4. **Calendar buckets for cons-loss.** Exit timestamps in ms vs seconds vs bar index are normalized in `note_closed_trade_day`.
5. **Compile path.** Object-mode uses `CompileStrategyBroker` —aligned fill rules, smaller surface than full interpreter metrics. Prefer interpret mode for full `strategy.closedtrades.*` analytics unless compile coverage is verified.

Worked examples

Market long / close

```
//@version=6
strategy("Demo", overlay=true, initial_capital=100000)
if bar_index == 5
    strategy.entry("L", strategy.long, qty=10)
if bar_index == 10
    strategy.close("L", qty_percent=100)
```

Parity fixtures under `tests/fixtures/parity/pine/` encode expected event sequences for such scripts.

Pending stop entry

```
//@version=6
strategy("Breakout", overlay=true)
strategy.entry("Break", strategy.long, stop=high[1])
```

Creates a pending stop; subsequent bars' `process_pending_orders` fill when `high` trades through the stop, using open-gap-aware fill prices.



OCA bracket sketch

```
//@version=6
strategy("Bracket", overlay=true)
// after entry "L" is open:
strategy.order("TP", strategy.short, qty=1, limit=tp, oca_name="br", oca_type=strategy.oca.cancel)
strategy.order("SL", strategy.short, qty=1, stop=sl, oca_name="br", oca_type=strategy.oca.cancel)
```

First fill cancels the sibling.

Failure modes

Symptom	Cause
Entry never fills	Pending stop/limit; OHLC never trades through; or risk block
Double entries	Pyramiding > 0 or missing close
Events missing <code>script_id</code>	Host forgot to stamp after <code>drain_events</code> (Runtime does this)
PnL off vs TV	Commission type, slippage ticks, mintick, or fill price model
Interpret vs compile event drift	Broker subset / timing—diff with <code>tests/test_compiler_strategy.py</code>

See also

- [Events](#)
- [Strategy broker \(compile\)](#)
- [pyne-worker](#)
- [Pro API backtest](#)

COLLECTIONS

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title: "Collections" description: "array., *matrix.*, and map.* runtime semantics, method dispatch, and type tags."

COLLECTIONS

Abstract

Pine collections—**arrays**, **matrices**, and **maps**—are mutable values that live in the evaluator context (often under `var` so identity persists across bars). PYNE maps them to Python `list`, a dedicated `Matrix` type, and a `Map / dict`-like type respectively, with builtin functions and instance-method sugar (`a.push(x)` → `array.push(a, x)`).

Conceptual model

```

flowchart LR
    Pine[array / matrix / map API] --> Disp[Qualified builtins]
    Pine --> Meth[Instance method markers]
    Meth --> Disp
    Disp --> List[list]
    Disp --> Mat[Matrix]
    Disp --> MapT[Map / dict]

```

Interface surface

Arrays (`array.*`)

Represented as Python lists. Highlights from the dispatch map:

Category	Functions
Construct	<code>array.new</code> , <code>array.new_<type></code> , <code>array.from</code>
Mutate	<code>push</code> , <code>pop</code> , <code>shift</code> , <code>unshift</code> , <code>insert</code> , <code>remove</code> , <code>set</code> , <code>fill</code> , <code>clear</code> , <code>reverse</code> , <code>sort</code>
Access	<code>get</code> , <code>first</code> , <code>last</code> , <code>size</code> , <code>includes</code> , <code>indexof</code> , <code>lastindexof</code> , <code>slice</code>
Aggregate	<code>sum</code> , <code>avg</code> , <code>min</code> , <code>max</code> , <code>median</code> , <code>mode</code> , <code>range</code> , <code>stdev</code> , <code>variance</code> , <code>covariance</code> , <code>percentiles</code> , <code>percentrank</code>
Search	<code>binary_search</code> , <code>binary_search_leftmost</code> , <code>binary_search_rightmost</code> — UDT arrays take <code>sort_field</code> (const int index, default 0, or const string name; array must be sorted by that field ascending)
Higher-order	<code>every</code> , <code>some</code> , <code>copy</code> , <code>concat</code> , <code>join</code> , <code>abs</code>

Indexing: Pine array indices are **0-based from the start of the list** (unlike series history). Negative indices are rejected for series; array paths use list semantics consistent with the builtin handlers.

UDT `sort_field` (0.3.6 / Pine August 2026): `array.sort`, `array.sort_indices`, and `array.binary_search*` compare one field of user-defined type elements. Pass a const int field index (`0` is the first declared field, and the default for search) or a const string field name. Binary search requires the array to already be sorted by that field in ascending order. Method form (`a.binary_search(value, "id")`) and `sort_field=` kwargs work on interpret and compile.

Series-like values passed into array ops unwrap via `.history` (reversed to chronological) when duck-typed.

Matrices (`matrix.*`)

`Matrix` supports:

- Construction: `matrix.new`
- Element access: `matrix.get` / `set`, and `m[row, col]` via subscript
- Shape: `rows`, `columns`, `elements_count`
- Row/column ops: add/remove/copy, sum/avg/min/max/mode, fill
- Aggregates: sum/avg/min/max/mode all
- Transforms: `transpose`, fill diagonal, and further linear helpers in the evaluator map

Method sugar: `m.rows()` → `matrix.rows(m)`.



Maps (`map.*`)

Function	Role
<code>map.new</code>	Empty map
<code>put</code> / <code>put_all</code> / <code>get</code> / <code>remove</code> / <code>clear</code>	Mutation
<code>contains</code> / <code>keys</code> / <code>values</code> / <code>size</code> / <code>copy</code>	Query

Keys follow Pine map rules as implemented (hashable Python keys). Compile object-mode lowers maps to ordinary Python `dict`s.

Internals

Path	Role
<code>src/pynescrypt/ast/evaluator/builtins/arrays.py</code>	Array builtins
<code>src/pynescrypt/ast/evaluator/builtins/matrix.py</code>	<code>Matrix</code> type
<code>src/pynescrypt/ast/evaluator/builtins/matrix_evaluator.py</code>	<code>matrix.*</code> handlers
<code>src/pynescrypt/ast/evaluator/builtins/map.py</code>	<code>Map</code> type
<code>src/pynescrypt/ast/evaluator/builtins/map_evaluator.py</code>	<code>map.*</code> handlers
<code>src/pynescrypt/ast/evaluator/names.py</code>	Method markers for list/Matrix
<code>tests/test_collections*.py</code> , <code>test_map_collections.py</code> , <code>test_matrix_*.py</code>	Coverage

Multi-dispatch type tags (`array.float` , `matrix.string` , ...) interact with method overloading and `na` exclusion lists so `na`-typed optionals do not bind to collection `tostring` overloads.

Invariants & edge cases

- Identity under `var`.** `var a = array.new_float(0)` keeps one list across bars; reassigning `a := array.new_float(0)` replaces it.
- Drawing-typed arrays.** `array.new_label` etc. store drawing object references; delete semantics remain drawing-registry concerns.
- Empty array overload match.** Empty arrays match any `array.*` element tag in dispatch (no sample element).
- Matrix unpack hazard.** `StatementEvaluator` refuses to treat matrices as general iterables for tuple unpack—prevents corrupting multi-assign from row iteration.
- Compile mode.** Maps/UDTs force **object mode**; pure array numeric work may still numeric-compile depending on visitor detection—verify generated code when performance-critical.



Worked examples

Rolling buffer

```
//@version=6
indicator("buf")
var floats = array.new_float(0)
array.push(floats, close)
if array.size(floats) > 20
    array.shift(floats)
plot(array.avg(floats))
```

Matrix element

```
var m = matrix.new<float>(2, 2, 0.0)
matrix.set(m, 0, 1, close)
plot(matrix.get(m, 0, 1))
```

Map counter

```
var counts = map.new<string, int>()
map.put(counts, "n", nz(map.get(counts, "n")) + 1)
```

Binary search on a UDT array

```
//@version=6
indicator("udt search")
type Item
    int id
    float v
var a = array.new<Item>()
if bar_index == 0
    array.push(a, Item.new(2, 20.0))
    array.push(a, Item.new(1, 10.0))
    array.push(a, Item.new(3, 30.0))
    array.sort(a, order.ascending, "id")
plot(array.binary_search(a, 2, "id"))
plot(a.binary_search_leftmost(2, sort_field="id"))
```

Failure modes

Symptom	Cause
<code>map.get</code> requires map and key	Wrong arity or non-Map receiver
Index errors	OOB get/set without Pine <code>na</code> guard in that handler
Method not found	Typo or non-list receiver after series unwrap failed
Shared mutation across runs	Reused evaluator context without reset



See also

- [Expressions & statements](#)
- [Drawing & plotting](#) — arrays of labels/lines
- [Compiler object mode](#)

DRAWING PLOTTING

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title: "Drawing and plotting" description: "plot* side effects, DrawingRegistry objects, and export shapes for hosts and AXIS."

DRAWING AND PLOTTING

Abstract

Visual effects in Pine are **side channels**: they do not change pure arithmetic results, but they are first-class runtime outputs. PYNE records plots in `PlotRegistry` and drawing primitives (lines, boxes, labels, tables, polylines, linefills) in `DrawingRegistry`, then serializes them for the Pro API and optional AXIS. Evaluation never requires a UI—tests assert on registries alone.



Conceptual model

```
flowchart TB
    Script[plot / line.new / label.new] --> Reg[Registries]
    Reg --> PlotR[PlotRegistry]
    Reg --> DrawR[DrawingRegistry]
    PlotR --> Host[Host series envelope]
    DrawR --> API[export_for_api]
    API --> AXIS[AXIS / chart clients]
```

Interface surface

Plotting (`PlottingFunctionsMixin`)

Function	Kind	Notes
<code>plot</code>	<code>plot</code>	Returns <code>Plot</code> id for <code>fill(plot1, plot2)</code> ; series values in host export
<code>plotshape</code> / <code>plotchar</code> / <code>plotarrow</code>	markers	Condition series + <code>plot_meta.kind</code> (style/location/char). Visuals are not in <code>drawings</code>
<code>barcolor</code> / <code>plotbar</code> / <code>plotcandle</code>	overlays	Per-bar color / OHLC series + kind stamp (0.3.12 dual-host)
<code>hline</code>	horizontal	Both hosts export a constant price series keyed by title + <code>plot_meta.kind: "hline"</code> . Not in <code>drawings</code>
<code>bgcolor</code> / <code>barcolor</code>	coloring	Color-string series + <code>plot_meta.kind</code> . Not in <code>drawings</code>
<code>fill</code>	fill	Both export a titled series key (values often all-null); band color + plot refs live in <code>plot_meta</code>

Style constants: `plot.linestyle_solid` / `dashed` / `dotted` . Plot style tokens: `plot.style_line` , `linebr` , `stepline` , `steplinebr` , `stepline_diamond` , `histogram` , `columns` , `cross` , `area` , `areabr` , `circles` .

Each call constructs a `Plot` dataclass (`kind` , `series` , `title` , `color` , `linewidth` , text formatting, `force_overlay` , ...) and appends it to `PlotRegistry.plots` .

Hosts often **also** capture per-bar plot values on the evaluator (`plot_outputs` / `_plot_value_cols`) for time-aligned series maps—see `src/pynescript/runtime/evaluator.py` and packaging in

`src/pynescript/runtime/host.py` (`backend.evaluator` / `backend.runtime` re-export those modules).

`PYNE_LIGHT_PLOTS=1` skips columnar capture, plot-registry handles, and `input.*` metadata (corpus OK/fail only).

Title defaults and series keys

Runtime packages plots into `result["series"]` / `result["plot_meta"]` on **both** hosts. `result["drawings"]` is **geometry-only** (line / box / label / polyline / table / linefill). Compile `__drawings` is an internal event list that Runtime **folds and strips** before JSON — hosts never need it. Empty or missing titles get defaults; collisions get `_2` , `_3` , ... suffixes so map keys stay unique.



Call	Default title	Notes
<code>plot</code>	<code>plot_0</code> , <code>plot_1</code> , ...	Compile emits <code>plot_{n}</code> at emit time; interpret empty title → <code>plot_{index}</code> at JSON packaging
<code>hline</code>	<code>hline</code>	Uniquified <code>hline</code> , <code>hline_2</code> , ... when untitled levels repeat
<code>fill</code>	<code>fill</code>	Same uniquify rule (<code>Background</code> / <code>Background_2</code> , ...)
<code>bgcolor</code>	<code>bgcolor</code>	Series + <code>plot_meta</code> (both hosts)
<code>plotshape</code>	<code>shape</code>	Series + <code>plot_meta</code> (both hosts)

Explicit `title=` (or positional title) wins when present.

`hline` — constant series (both modes)

Interpret (`src/pynescrypt/runtime/evaluator.py` + Runtime packaging):

- Captures kind `hline` with the price as the series cell each bar.
- JSON export **forward-fills** nulls with the known constant so AXIS can draw a full-width level.
- `plot_meta[title]` includes `kind: "hline"` , `price` , `color` , `linewidth` , ...

Compile:

- Emits a **plot array** (`safe_float(price)` every bar) under the uniquified title and stamps `plot_meta.kind: "hline"` .
- May also append an internal `__drawings` event; Runtime packaging **strips** visual kinds so host `drawings` stays geometry-only.
- Host wraps NaN → null; constant levels remain numeric across the bar range.

Parity tests: `tests/test_multi_plot_cross.py` (`test_hline_series_keys_compile_matches_interpret`),
`tests/test_plot_drawing_dual_host.py` , `tests/test_compiler_objects.py` .

`fill` — titled series key for key-set parity (both modes)

Interpret:

- Meta stores sibling series titles as `plot1` / `plot2` (resolved from `plot()` handles).
- Capture may hold a color string on the column; **Runtime JSON packaging** typically emits **all-null** series cells for fills (non-numeric / non-bgcolor strings → `null`). Band **color** and plot refs stay in `plot_meta` (and registry fill objects when ids are enabled).

Compile:

- Registers a series key with expression `None` (float64 NaN → JSON `null`) under the uniquified title so **series key sets match interpret**.
- Band details (plot refs, color) land in `plot_meta` . An internal `__drawings fill` event, if emitted, is **stripped** before Runtime JSON.

Parity tests: `test_fill_background_series_keys_compile_matches_interpret` ,
`test_fill_title_series_key_and_drawings` , `tests/test_plot_drawing_dual_host.py` .



Import stubs must not leak as series strings

Unresolved `import ...` libraries become chainable stubs marked `__pine_import_stub__ = True` (repr like `<PineImportStub namespace/name/version>`). If a script plots a stub member, Runtime's `_json_plot_value` maps those cells to `null`—never the stub repr or a color-like string—so AXIS and interpret/compile comparisons never see `"<PineImportStub ...>"` as a plot series value.

Drawing objects (`DrawingBuiltinsMixin`)

Object types: `Line`, `Box`, `Label`, `Table`, `Polyline`, `LineFill`, `ChartPoint`.

Typical factories: `line.new`, `box.new`, `label.new`, `table.new`, `polyline.new`, `linefill.new`, plus getters/setters/deletes and `*.all` / last-bar helpers where implemented.

Instance methods resolve through namespace markers:

```
la.get_text() → label.get_text(la)
```

`DrawingRegistry.export_for_api(bar_times)` maps `xloc=bar_index` coordinates to wall times for chart clients, normalizes colors, and skips deleted objects. `xloc.bar_index` **past the last bar** (classic `bar_index + 1` on `barstate.islast`) is extrapolated from the series period (0.3.5); empty `bar_times` passes the bare bar index.

Line / box / label payloads include `force_overlay` for AXIS pane routing (0.3.6). `linefill.new` serializes as `type: "linefill"` quads (`t1 / p1 ... t4 / p4` from the two line endpoints, plus `color` / `bgcolor`).

Registry lifecycle

```
DrawingRegistry.reset() # clears drawings + PlotRegistry
```

Hosts must reset at run start so labels from a previous evaluation do not leak.

Drawing GC (`max*_count`)

Interpret path enforces TradingView®-style **garbage collection** on drawing factories (`line.new`, `label.new`, `box.new`, `polyline.new`, ...):

Declaration kwargs (<code>indicator</code> / <code>strategy</code>)	Default	Hard cap
<code>max_lines_count</code>	50	500
<code>max_labels_count</code>	50	500
<code>max_boxes_count</code>	50	500
<code>max_polylines_count</code>	50	100

When more **active** objects of a type exist than the cap, the **oldest** are marked `deleted=True` so they leave `*.all` and `DrawingRegistry.export_for_api`. Caps are applied from the script declaration at run start (`DrawingRegistry.configure_from_declaration`); shrinking a cap immediately collects.

```
//@version=6
indicator("labels", overlay=true, max_labels_count=100)
if barstate.islast
    label.new(bar_index, high, "x")
```



Compile object-mode drawings are a simplified event list—GC caps apply on the **interpret** registry path; prefer interpret when you need full last-bar / `*.all` / cap behavior.

fill + plot_meta for AXIS

For dual `plot` + `fill` bands:

- 1. Each `plot()` contributes a **numeric series** under its title (default `plot_0`, ...).
- 2. `fill(p1, p2, ...)` contributes a **titled series key** (often all-null cells) so key-sets stay stable, plus `plot_meta` that carries band color and plot refs.
- 3. Hosts such as AXIS should read `plot_meta[title]` for `kind`, colors, and sibling plot titles (`plot1` / `plot2`) rather than treating fill series as prices.

Interpret packaging lives in `src/pynescrypt/runtime/host.py` / `evaluator.py` ; compile emits matching series keys. Internal `__drawings` fill events are folded/stripped before Runtime JSON.

Internals

Path	Role
<code>src/pynescrypt/ast/evaluator/builtins/plotting.py</code>	<code>Plot</code> , <code>PlotRegistry</code> , <code>plot*</code> handlers
<code>src/pynescrypt/ast/evaluator/builtins/drawing.py</code>	Drawing types, registry, builtins, <code>export_for_api</code> , <code>fold_compile_drawing_mutations</code>
<code>src/pynescrypt/runtime/evaluator.py</code>	Host plot capture: <code>hline</code> / <code>fill</code> / <code>bgcolor</code> / <code>plotshape</code> series + meta
<code>src/pynescrypt/runtime/host.py</code>	JSON packaging: series keys, stub nulling, <code>hline</code> fill-forward, fill plot refs
<code>src/pynescrypt/compiler/compiler.py</code>	<code>hline</code> / <code>fill</code> series + internal <code>__drawings</code> ; visual events lifted then stripped
<code>src/pynescrypt/ast/evaluator/names.py</code>	<code>_DRAWING_METHOD_NS</code>
<code>tests/test_plotting_effects.py</code> , <code>test_drawing_all_and_last_bar.py</code> , <code>test_multi_plot_cross.py</code> , <code>test_bgcolor_plotshape_export.py</code> , <code>test_plot_drawing_dual_host.py</code> , <code>test_compiler_objects.py</code>	Behavior / parity

Compile object-mode notes

Geometry factories and some marker-style calls force **object mode** (pure-Python bar loop with an append-only `__drawings` event list). Runtime packaging **lifts** visual events into `series` + `plot_meta.kind` , then **strips** them so host `drawings` is geometry-only — the same AXIS shape as interpret `export_for_api` .



Call	Host export (both modes)	Compile internals
<code>hline</code>	Constant series + <code>plot_meta.kind</code>	Series array; optional <code>__drawings</code> event stripped
<code>fill</code>	All-null series key + <code>plot_meta</code> color/refs	Series key; optional <code>__drawings</code> event stripped
<code>plotshape</code> / <code>plotchar</code> / <code>plotarrow</code>	Condition/signed series + <code>plot_meta.kind</code>	Visual events lifted into series, then stripped
<code>bgcolor</code> / <code>barcolor</code>	Color-string series + <code>plot_meta.kind</code>	Same lift/strip as other visuals
<code>plotbar</code> / <code>plotcandle</code>	OHLC series + kind stamp	Same lift/strip
<code>label</code> / <code>line</code> / <code>box</code> / <code>polyline</code> / <code>table</code> / <code>linefill</code>	Geometry objects in <code>drawings</code>	Handle dict + <code>__drawings</code> ; <code>set_*</code> / <code>*.delete</code> fold via <code>DrawingRegistry.fold_compile_drawing_mutations</code>

Numeric mode still supports plain `plot` float64 arrays; geometry or marker calls can flip object mode for that script.

Invariants & edge cases

1. **`plot` returns an id.** Required for `fill`; treating it as “void” breaks band fills. Hosts that skip `PlotRegistry` unless `fill()` appears in source must still soft-fail fill gracefully.
2. **Deleted flag.** Soft-delete keeps history for debugging; `active()` filters deleted plots and GC-collected drawings.
3. **`max_*_count` GC.** Oldest active drawings are deleted when caps are exceeded (interpret registry).
4. **Series in drawings.** Prices may be series wrappers—export coerces `.current` and maps NaN → omit.
5. **Compile path.** Numeric mode supports `plot` arrays; geometry/markers force object mode with an internal `__drawings` event list. Runtime **folds** `set_*` / `*.delete`, **lifts** visual events into `series` + `plot_meta.kind`, then **strips** them so `drawings` is geometry-only. Prefer interpret when you need full last-bar / `*.all` / registry GC — not because visuals lack series.
6. **Import stubs → null.** `__pine_import_stub__` and `"<PineImportStub...>"` strings never appear as plot series cells after Runtime packaging.
7. **No GPU/canvas in-process.** Registries are data; AXIS/HOOX render elsewhere.

Worked examples

Dual plot + fill

```
//@version=6
indicator("BB mid")
basis = ta.sma(close, 20)
p1 = plot(basis)
p2 = plot(basis * 1.01)
fill(p1, p2, color=color.new(color.blue, 80))
```



Label on last bar

```
//@version=6
indicator("lbl", overlay=true)
if barstate.islast
    label.new(bar_index, high, str.tostring(close))
```

Hosts read `DrawingRegistry.labels` or API export after the run.

Failure modes

Symptom	Cause
Empty drawings in API	Forgot <code>DrawingRegistry.reset</code> timing / export; or script never called <code>*.new</code>
fill no-ops	plot ids not retained from <code>plot()</code> returns
Stale labels across HTTP runs	Missing registry reset between <code>Runtime.run</code> calls (Runtime resets—custom hosts must)
Object mode missing drawing	Compile still folds <code>set_*</code> / <code>delete</code> ; leftover events usually mean fold did not run
Only 50 labels/lines visible	Default <code>max*_count</code> ; raise on <code>indicator</code> / <code>strategy</code> ($\leq$ hard cap)
fill series all null	Expected for band keys; use <code>plot_meta</code> for color + plot refs

See also

- [Builtins hub](#)
- [Pro API preview / chart renderer](#)
- [AXIS docs](#)
- [Compiler overview](#)

REQUEST INPUT

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title: "request.* and input.*" description: "Multi-symbol/timeframe requests, fundamental na semantics, host series bind, and input parameter resolution."

`request.*` **AND** `input.*`

Abstract

Two namespaces couple scripts to the **host environment**: `input.*` declares parameters (with defaults and UI metadata), and `request.*` pulls data from other symbols, timeframes, or fundamental/economic sources. PYNE implements both as builtins with explicit extension points—`data_feed`, `data_provider`, and `_input_overrides`—so the same script can run offline or online with real market data.

Design rule: missing multi-symbol or fundamental data yields `na`, never silent substitution of chart OHLCV (no close-as-dividend, no chart volume as UPVOL).



Conceptual model

```
flowchart TB
  subgraph inputs [Inputs]
    Def[defval in script]
    Over[_input_overrides by title]
    Def --> Res[Resolved value]
    Over --> Res
    Res --> Meta[_input_declarations]
  end
  end
  subgraph req [request.security]
    Call[symbol + expression] --> Chart{Chart symbol?}
    Chart -->|yes + simple OHLCV| Host[Chart series / provider]
    Chart -->|foreign + feed hit| Feed[data_feed / data_provider]
    Chart -->|foreign + no data| Na[na]
    Host --> Out[Series / scalar]
    Feed --> Out
    Na --> Out
  end
  end
```

Interface surface

input.* (InputBuiltinsMixin)

Builtin	Purpose
input	Generic defval + title/tooltip/inline/group/confirm/active
input.bool / int / float / string / color	Typed scalars
input.price / source / time / timeframe / session / symbol	Domain inputs
input.enum / input.text_area	Enumerations and multiline text

Runtime value: each call returns the resolved value (override if title matches, else default). Metadata is appended to `_input_declarations` for settings panels and LSP-adjacent hosts.

Overrides live on the evaluator as `_input_overrides: dict[title, value]`.

request.* (RequestBuiltinsMixin)

Builtin	Role
request.security	Other symbol / timeframe expression
request.security_lower_tf	Lower-TF array expansion
request.dividends / earnings / splits	Corporate actions
request.financial / economic / quandl	Fundamentals / external series
request.currency_rate	FX conversion helper
request.seed	Deterministic pseudo-series
request.footprint	Volume footprint object (v6 surface)



request.security — interpret path

Resolution order (simplified):

1. Normalize symbol (series → last element; `ticker.*` → symbol string) and timeframe.
2. Classify **chart vs foreign** (`_is_chart_symbol` vs host `syminfo` / provider symbol).
3. Try `data_feed.fetch_latest_ohlcv` / ticker, then `data_provider.fetch`.
4. **Same-symbol coarser timeframe (HTF resample, 0.3.4+)**: bucket chart OHLCV to the requested TF; **last completed** HTF bar by default (lookahead_off-style). Allowlisted simple TA on HTF series — `ta.sma` / `ta.ema` / `ta.rsi` / `ta.atr` / `ta.wma` / `ta.rma` (bare `close` / `high` / ... source + const length; `ta.atr(n)` length-only) — evaluates on the unique completed HTF series. Nested `ta.*`, UDFs, and non-const lengths stay `na`. `barmerge.gaps_on` / `lookahead_on` are honored on these resample paths (0.6.0, finalized-bucket lookahead in 0.6.1): gaps deliver the value only on bucket-start bars, lookahead leaks each HTF bucket's final value from the period start on historical bars. Both stay unused on passthrough / provider / complex-na paths. Policy meta is exposed on `result["meta"]["request_security"]` (`gaps_supported` / `lookahead_supported` , `gaps_applied` / `lookahead_applied`).
5. **Foreign + pre-evaluated expression** (UDF result, list/tuple of chart values, non-string expr) without multi-symbol data → `na` (not chart close as “dividends”).
6. Fundamental / non-equity prefixes (`DIVIDEND` , `FACTSET` , `EARNINGS` , `ESD_`) with no feed hit → `na` (no mock OHLCV).
7. Bare equity-style string names may still use **legacy mock** prices for offline demos when no feed is wired.

`ChartOHLCVProvider` (wired by `Runtime` via `resolve_request_sources` when unset) **only serves the chart ticker**. Foreign tickers get empty series, so interpret returns `na` unless a real multi-symbol feed is configured.

request.security — compile path

Compile lowering is intentionally narrow ([compiler overview](#)):

Case	Emit
Same-symbol (<code>syminfo.ticker</code> / empty chart id) and simple OHLCV expr (<code>close</code> , <code>high[1]</code> , ...)	Passthrough chart array sample
Foreign tickers (<code>UPVOL.NY</code> , <code>ESD_FACTSET</code> , ...)	<code>np.nan</code>
Complex third arg (UDFs, <code>year_sum(close)</code> , non-allowlisted <code>ta.*</code> , arithmetic)	<code>np.nan</code>
Other <code>request.*</code> APIs	<code>np.nan</code> (object mode)

No inventing chart-close-as-dividends or chart-volume-as-advance/decline series.

Footprint (`FootprintBuiltinsMixin`)

Types `Footprint` and `VolumeRow` expose buy/sell volume, delta, VAH/VAL/POC rows, and per-row imbalance helpers—aligned with the v6 footprint surface inventory.

Mode selection (`mode=auto`)

Pro API `/run` defaults body `mode` to **auto**. `Runtime._compile_eligible` **rejects compile** when the source contains `request.` (or top-level `import`), so auto **prefers interpret** for any script that uses `request.*`:

```
request.* present → compile_fallback_reason = "request.* not supported in compile path"
                    → auto_backend = interpret
```



Forced `mode=compile` still runs the narrow lowering above (simple same-symbol only; else `na`). Input overrides also force interpret under auto.

See [Compiler overview](#) for warm compile, numeric vs object mode, and eligibility.

Host series bind (interpret)

Assigning host OHLCV / time series into a **user name** never aliases the host buffer by reference.

`_bind_series_name` copies the current scalar into a fresh series for the user name so patterns like:

```
last_t := na(close) ? last_t[1] : time
```

cannot corrupt `time[j]` history. That bug previously broke TTM / `year_sum` -style windows (e.g. dividend-yield scripts).

Internals

Path	Role
<code>src/pynescrypt/ast/evaluator/builtins/input.py</code>	Input handlers + declarations
<code>src/pynescrypt/ast/evaluator/builtins/request.py</code>	<code>request.*</code> + chart/foreign / na policy
<code>src/pynescrypt/ast/evaluator/statements.py</code>	<code>_bind_series_name</code> (no OHLCV alias)
<code>src/pynescrypt/util/data.py</code>	<code>ChartOHLCVProvider</code> , <code>resolve_request_sources</code>
<code>src/pynescrypt/compiler/compiler.py</code>	Same-symbol OHLCV-only <code>security</code> lower
<code>src/pynescrypt/runtime/host.py</code>	Package Runtime SoT; feed wiring; <code>_compile_eligible</code> / <code>_run_auto</code> ; copies policy onto <code>meta.request_security</code>
<code>backend/runtime.py</code>	Compat re-export of package Runtime (not the implementation)
<code>tests/test_dividend_yield_parity.py</code> , <code>test_datafeed_wiring.py</code>	na parity + chart provider

Invariants & edge cases

1. **Inputs are pure values at runtime.** Titles matter only for override keys and UI metadata—not for Pine type identity.
2. **Foreign without data** → `na` . Prefer honest missing data over inventing chart series as multi-asset results.
3. **Chart provider is chart-only.** Multi-asset accuracy needs a real `data_feed` / `data_provider` .
4. **Mocks are limited.** Equity-style bare symbols may still mock offline; fundamental prefixes and foreign pre-evaluated exprs do not.
5. **Dynamic symbols.** List/series symbols resolve to the latest element—supports loops constructing ticker ids.
6. **Lower TF.** `request.security_lower_tf` returns array-like structures; length scales with simulated lower-TF density when mocking.
7. **`mode=auto` + `request.*` → interpret.** Compile path is not a full multi-asset substitute (`_compile_eligible` rejects `request.`).

8. **HTF is last-completed only.** Forming HTF buckets are never returned; `gaps` / `lookahead` kwargs do not change that.

Worked examples

Parameterized length

```
//@version=6
indicator("len")
len = input.int(14, "Length", minval=1)
plot(ta.sma(close, len))
```

Host:

```
ev._input_overrides = {"Length": 21}
```

Multi-timeframe close / simple HTF TA (chart symbol)

```
//@version=6
indicator("HTF")
htf = request.security(syminfo.tickerid, "D", close)
htf_sma = request.security(syminfo.tickerid, "D", ta.sma(close, 20))
plot(htf)
plot(htf_sma)
```

Same-symbol coarser TF: interpret buckets chart OHLCV (last **completed** HTF bar) and evaluates allowlisted simple `ta.*` (`sma` / `ema` / `rsi` / `atr`) on that series. Compile may passthrough simple same-symbol OHLCV; complex foreign/security still → `na`. Inspect `meta.request_security.policies` (`htf_ohlcw_resample`, `htf_simple_ta_resample`, `foreign_na`, `complex_htf_na`, ...).

Intentional `na` — dividend yield (fundamentals missing)

```
//@version=6
indicator("div")
year_sum(src) =>
    ta.cum(src)
div_ticker = ticker.new("ESD_FACTSET", "X;Y;DIVIDENDS")
div_ttm = request.security(div_ticker, "D", year_sum(close), barmerge.gaps_on, lookahead=barmerge.L
plot(div_ttm)
```

Without a fundamentals feed: **interpret and compile both plot `na` for `div_ttm`** —not chart close as fake TTM dividends. Covered by `tests/test_dividend_yield_parity.py`.

Intentional `na` — CVI / UPVOL-style foreign OHLCV

```
//@version=6
indicator("cvi")
up = request.security("UPVOL.NY", "D", close)
plot(up)
```



Foreign ticker + no multi-symbol data → `na` . Compile never rewrites this as chart close; inventing advance/decline volume from the host series is forbidden.

Failure modes

Symptom	Cause
All- <code>na</code> multi-asset plots	Foreign symbol / fundamental expr; no feed (expected)
Always ~100 mock prices	Bare equity string + no feed; legacy mock path
Input override ignored	Title string mismatch (including empty title)
Footprint fields zero	<code>request.footprint</code> without configured footprint data
<code>time[1]</code> / history wrong after assign	Should be fixed: host bind no longer aliases OHLCV
Lookahead surprises	Host data alignment / gaps—not automatic TV replay guarantees
<code>auto_backend=interpret</code> with <code>request.*</code>	Eligibility prefilter; use interpret or wire multi-symbol feed

See also

- [Compiler overview](#) — auto eligibility, same-symbol security lower
- [Builtins hub](#)
- [Series & history](#)
- [Pro API run endpoint](#)
- [Runtime hub](#)

LIBRARIES

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title: "Libraries" description: "In-process library export/import, LibraryRegistry, and evaluate-order requirements."

LIBRARIES

Abstract

TradingView libraries publish as `namespace/Name/version` with `export` ed members. PYNE implements an **in-process** analogue: evaluate a `library("Title")` script (or register source by path), collect exports (`export const` , exported functions, types), then `import` binds an alias to a `LibraryModule` for `alias.member` access. There is no network publish step—the registry is the host’s responsibility.

Conceptual model

```

flowchart LR
  LibSrc[library script] --> Eval[Evaluate once]
  Eval --> Exp[pending exports]
  Exp --> Reg[LibraryRegistry]
  Imp[import ns/Name/version as alias] --> Reg
  Reg --> Mod[LibraryModule]
  Mod --> Use[alias.member in consumer]

```

Interface surface

Library script

```

//@version=6
library("MyMath")

a + b

```

On script completion, `StatementEvaluator` finalizes `_pending_library_exports` onto the active `LibraryModule` and calls `LibraryRegistry.register`.

Consumer import

```

//@version=6
indicator("use")

plot(M.add(close, M.PHI))

```

Resolution (`LibraryRegistry.lookup`):

1. Exact (`namespace, name, version`) if provided and registered.
2. Else title-only match (`MyMath`) for local evaluation workflows without publisher path.

API on the evaluator

```

ev.register_library_source(namespace, name, version, source_str)
mod = ev.lookup_library(namespace=..., name=..., version=...)

```

`register_library_source` stores Pine text for **lazy** load on the first matching `import`. `_load_library_source` evaluates definitions only (skips showcase body) and **finalizes** `_pending_library_exports` onto the `LibraryModule` so `export const` / exported functions bind (0.3.8). Evaluating a library script on the same evaluator still populates exports eagerly.

Host evaluate (0.3.7+)

Hosts pass published sources without a pre-eval step:



```
Runtime(symbol="TEST").run(
    consumer_source,
    ohlcv,
    mode="interpret", # or "auto" – import is compile-ineligible
    libraries=[
        {"namespace": "ns", "name": "Lib", "version": 1, "source": library_source},
    ],
)
```

The same list is accepted on Pro API and **pyne-worker** `POST /run` as `libraries`. `pyne-worker` also stores the array on deployed scripts so `cron / script_id` reuse it. Pro API slices to **32** entries (`backend/app.py`); `Runtime.run` itself does not cap. `mode=compile` has no import path — `mode=auto` treats `import` as compile-ineligible and **forwards `libraries=` into `interpret` fallback** (0.3.9).

LibraryModule

Dataclass with `title`, optional `namespace / version`, and `exports: dict[str, Any]`. Attribute access reads exports; missing members raise `AttributeError` with a clear message. `NameEvaluator` routes `alias.member` when the base value is a `LibraryModule`.

Internals

Path	Role
<code>src/pynescrypt/ast/evaluator/libraries.py</code>	<code>LibraryModule</code> , <code>LibraryRegistry</code>
<code>src/pynescrypt/ast/evaluator/statements.py</code>	export registration, import visit, library finalize
<code>src/pynescrypt/ast/evaluator/base.py</code>	registry fields on <code>BaseEvaluator</code>
<code>src/pynescrypt/ast/evaluator/__init__.py</code>	<code>register_library_source</code> , <code>lookup_library</code>
<code>src/pynescrypt/runtime/host.py</code>	<code>Runtime.run(..., libraries=)</code> + auto-mode forward
<code>tests/test_library_export_import.py</code>	Round-trip coverage

Exportable values include callables (user functions), constants, and type-related bindings depending on declaration paths. Exported methods participate in the same call machinery as local functions once retrieved from the module.

Invariants & edge cases

- 1. **Evaluate library before consumer** (or register source for lazy import). Empty registry → import failure.
- 2. **Same evaluator instance** (or shared registry) must see both scripts—registries are not process-global unless the host shares them.
- 3. **Version is part of the path key**. Mismatched version does not fall back to another version automatically when namespace is specified.
- 4. **Title registration** always updates `_by_title` so simple local titles work without publisher metadata.
- 5. **No TV account auth**. Publishing/versioning outside the process is out of scope.



Worked examples

Explicit registration

```
from pynescrypt.ast.evaluator import NodeLiteralEvaluator

ev = NodeLiteralEvaluator()
ev.evaluate_script(library_source)    # registers by title
ev.evaluate_script(consumer_source)  # import by title or path
```

Path registration without pre-eval

```
ev.register_library_source("user", "MyMath", 1, library_source)
# import user/MyMath/1 loads definitions and finalizes exports on first visit
```

Failure modes

Symptom	Cause
Import resolves to nothing	Library never registered / wrong title
Library 'X' has no exported member 'y'	Missing <code>export</code> or typo
Stale exports	Re-evaluated library without re-import; host cache
Version conflict	Multiple libs same title—last <code>register</code> wins for title key

See also

- [Expressions & statements](#)
- [Builtins declarations](#)
- [End-user library API guide](#)

EVENTS

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title: "Strategy events" description: "StrategyEvent shape, emission points, parity corpus, and host serialization."

STRATEGY EVENTS

Abstract

A **strategy event** is the immutable, structured record of one `strategy.*` action (or fill side-effect) at a specific bar. Events are the wire format between PYNE's Python evaluator, **PyneTS** (`StrategyEvent` on `RuntimeResult`), and downstream HOOX trade workers (via **pyne-worker**). Changing the dataclass without updating the TS library and tests is a contract break.



Conceptual model

```
flowchart LR
    Builtin[strategy.entry / order / ...] --> Emit[Construct StrategyEvent]
    Fill[process_pending_orders] --> Emit
    Emit --> Buf[StrategyState._events]
    Host[Bar loop] --> Drain[drain_events]
    Buf --> Drain
    Drain --> Dict[to_dict + script_id/run_id]
    Dict --> TS[PyneTS RuntimeResult.events]
    Dict --> HOOX[pyne-worker → trade-worker]
```

Interface surface

StrategyEvent fields

Defined in `src/pynescrypt/ast/evaluator/events.py`:

Field	Type (logical)	Notes
kind	entry exit close close_all cancel cancel_all order	Discriminator
id	str None	Order / entry id
direction	long short None	
qty	float None	
order_type	market limit stop None	Stop-limit may appear as composite handling elsewhere
limit / stop	float None	Prices when relevant
oca_name	str None	OCA group
comment	str None	Includes system comments (<code>risk_blocked</code> , <code>fill:...</code>)
bar_index	int	From context at emit
bar_time	int	From context
ohlc	4-tuple → list in JSON	Bar snapshot
script_id	str	Host-stamped (source hash)
run_id	str	Host-stamped run uuid

Frozen dataclass → `to_dict()` always includes keys (JSON nulls for unspecified fields) so parity diffs are stable.

Emission sources

1. **Explicit builtins** — `strategy.entry` , `exit` , `close` , `close_all` , `cancel` , `cancel_all` , `order` push events as they execute.
2. **Broker fills** — `process_pending_orders` emits fill-related `order` / `entry` / `cancel` (OCA) events.
3. **Risk gates** — blocked entries may emit with `comment="risk_blocked"` .



Host collection pattern

```
evaluator.reset_events()
# ... process_pending_orders + visit(tree)
for ev in evaluator._strategy_state.drain_events():
    d = ev.to_dict()
    d["script_id"] = script_id
    d["run_id"] = run_id
    all_events.append(d)
```

`Runtime.run` implements this and returns events in the response envelope.

Compile path

`CompileStrategyBroker` accumulates plain dict events (`__events` in the return mapping) with the same field names for kinds it supports. Exit/trail/OCA/risk halt comments (`risk_blocked`) are on both hosts for the 0.3.4+ surface; remaining metric accessors still trail the interpreter.

Internals

Path	Role
<code>src/pynescrypt/ast/evaluator/events.py</code>	<code>StrategyEvent</code>
<code>src/pynescrypt/ast/evaluator/builtins/strategy.py</code>	Emit sites + <code>drain_events</code>
<code>src/pynescrypt/compiler/strategy_broker.py</code>	Compile-mode <code>_emit</code>
<code>tests/test_strategy_events.py</code>	Unit behavior
<code>tests/test_parity.py</code> + <code>tests/fixtures/parity/</code>	Cross-port oracle
<code>pynets/src/runtime/interpret.ts</code> <code>StrategyEvent</code>	TS library twin

Parity corpus scripts (`strategy_01_entry_long.pine` , ...) regenerate expected JSON via:

```
python tests/fixtures/parity/generate_fixtures.py
```

Tests strip `script_id` / `run_id` before compare.

Invariants & edge cases

1. **Immutability.** Do not mutate events after construction; drain copies and clears the buffer.
2. **Per-bar reset.** `reset_events` / drain prevents cross-bar leakage in tests that share evaluators.
3. **OHLC list vs tuple.** `to_dict` converts to list for JSON round-trips.
4. **Kind vocabulary is closed.** New kinds require TS + fixtures + this doc.
5. **Not alerts.** `alert()` / `alertcondition()` use a separate `alerts` channel—not `StrategyEvent` .

Worked examples

Expected single entry

Script enters long on a fixed bar; fixture JSON lists one event:



```
{
  "kind": "entry",
  "id": "L",
  "direction": "long",
  "qty": 1.0,
  "order_type": "market",
  "bar_index": 5
}
```

(plus nullables and ohlc—see real fixtures for full keys).

Cancel after OCA

Fill of one OCA sibling yields `cancel` events for others with `comment` reflecting `oca_cancel` / `oca_reduce`.

Failure modes

Symptom	Cause
Empty events with live strategy	Forgot drain; or strategy never called
Parity flake on ids	Comparing <code>script_id</code> / <code>run_id</code>
TS/Python mismatch	Field rename without dual update
Duplicate events	Not clearing buffer; double <code>process_pending_orders</code>

See also

- [Strategy builtins](#)
- [Strategy broker](#)
- [PyneTS](#)
- [pyne-worker](#)
- [HOOX live trading](#)
- [HOOX docs](#)

ALERTS

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title: "Alerts" description: "alert() / alertcondition() engine, host export, and L2 HTTP webhooks (Pro API + pyne-worker)."

ALERTS

Abstract

Pine `alert()` and `alertcondition()` fire into a **host side-channel**, not **strategy events**. The evaluator records structured firings with TradingView-style frequency rules. Hosts (Pro API and **pyne-worker**) export them on evaluate responses and can **POST** them to HTTP webhooks (roadmap **L2**).



Conceptual model

```
flowchart LR
    P[Pine alert / alertcondition] --> E[AlertsMixin]
    E --> L[_triggered_alerts]
    L --> R[Runtime.export]
    R --> API[API["/run alerts[]"]]
    R --> WH[HTTP webhook batch]
```

Layer	Module
Engine	src/pynescrypt/ast/evaluator/builtins/alerts.py (export_alerts / export_alerts_from_evaluator)
Package host	src/pynescrypt/runtime/host.py (clear_alerts per run, alerts on the envelope)
Pro API webhook	backend/alert_forwarder.py (backend/runtime.py is only a Runtime re-export)
Edge host	sibling pyne-worker alert_engine.py + alert_forwarder.py

Pine surface

alert(message, freq)

- Default `freq`: `alert.freq_once_per_bar` (canonical token `once_per_bar`).
- Constants (also bare strings): `alert.freq_once_per_bar`, `alert.freq_once_per_bar_close`, `alert.freq_all`.
- **once_per_bar** — first fire per bar for the same (source, title, message, freq) key.
- **once_per_bar_close** — only when the host marks the bar confirmed (`barstate.isconfirmed` / `islast` ; hosts without barstate treat bars as confirmed).
- **all** — every call.

alertcondition(condition, title, message)

- Records each evaluation in `alert_conditions` (debug / UI).
- When `condition` is true, also **fires** a host alert with `source: "alertcondition"` and once-per-bar semantics.

Event shape (to_dict / API)

```
{
  "message": "cross up",
  "freq": "once_per_bar",
  "bar_index": 42,
  "time": 1700000000000,
  "title": "optional",
  "source": "alert"
}
```

`source` is `"alert"` or `"alertcondition"`. Hosts stamp `script_id` / `run_id` (and edge cron may add `symbol` / `timeframe`).



Host export

Pro API (`POST /run`)

- Interpret path: full history of firings in `alerts` (and optional `alert_conditions`).
- Compile path: `alerts: []` today (alert side effects are interpret-only).
- See [POST /run](#).

pyne-worker

- `POST /run` returns the same `alerts` array.
- Cron filters to the **last closed bar** before webhook delivery so historical bars do not re-fire every minute.
- Health: `features.alerts` , `features.alert_webhooks` .

L2 webhooks

Both hosts POST JSON to an HTTPS endpoint.

Host	Default URL	Per-request / per-job
Pro API	env <code>ALERT_WEBHOOK_URL</code>	body <code>webhook_url</code>
pyne-worker	secret/var <code>ALERT_WEBHOOK_URL</code>	script/job <code>webhook_url</code>

Pro API request flags

Field	Default	Meaning
<code>webhook_url</code>	<code>" "</code>	Override destination
<code>forward_alerts</code>	<code>true</code>	Skip POST when false
<code>alert_last_bar</code>	<code>true</code>	Only firings on the last OHLCV bar
<code>alert_batch</code>	<code>true</code>	One batch body vs one POST per alert

Optional timeout: env `ALERT_WEBHOOK_TIMEOUT` (seconds, default 10).



Batch body

```
{
  "type": "pine_alert_batch",
  "source": "pyne-pro-api",
  "count": 1,
  "content": "cross up",
  "alerts": [
    {
      "type": "pine_alert",
      "source": "pyne-pro-api",
      "message": "cross up",
      "freq": "once_per_bar",
      "alert_source": "alert",
      "bar_index": 42,
      "time": 1700000000000
    }
  ]
}
```

Edge worker uses `"source": "pyne-worker"`. Discord-friendly `content` is set when a message is present.

Response meta

Successful `/run` may include:

```
{
  "alert_forward": {
    "forwarded": 1,
    "failed": 0,
    "filter": "last_bar",
    "url": "https://hooks.example.com/pine",
    "batch": true,
    "count": 1
  }
}
```

Webhook failures never fail the evaluate status; inspect `alert_forward` / `alert_forward_error`.

Worked example

```
curl -sS -X POST http://127.0.0.1:5002/run \
-H 'Content-Type: application/json' \
-d '{
  "script": "//@version=6\nindicator(\"a\")\nif bar_index == last_bar_index\n  alert(\"fire\")\n  "mode": "interpret",
  "webhook_url": "https://hooks.example.com/pine",
  "data": [
    {"time": 1, "open": 1, "high": 2, "low": 0.5, "close": 1.5, "volume": 10},
    {"time": 2, "open": 1.5, "high": 2.5, "low": 1.0, "close": 2.0, "volume": 12}
  ]
}'
```

Invariants

1. **Not strategy events** — trade mesh uses `events[]`; alerts are independent.



2. **Per-run clear** — hosts call `clear_alerts()` so runs do not leak firings.
3. **Last-bar default for webhooks** — full-history still available in the `alerts` response field.
4. **Compile path** — prefer `mode: "interpret"` (or auto fallback) when you need alerts.

See also

- [Strategy events](#)
- [POST /run](#)
- [Pro API usage](#)
- [Runtime bridge](#)
- [Roadmap \(L2 !\[\]\(b414d9e20ee1c1fb26c9f9f0f376be72_img.jpg\)](#))

OVERVIEW

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title: "Compiler overview" description: "Source-to-source compilation: CompilerVisitor, numeric vs object mode, run(time=...), and engine API."

COMPILER OVERVIEW

Abstract

The compile path lowers Pine's ASDL AST to **executable Python** that walks bars with an explicit index (`__bar_idx`) over contiguous `numpy` arrays—avoiding per-node visitor overhead. Pure numeric scripts wrap the loop in `@numba.njit`; scripts that touch UDTs, maps, drawings, or strategy select **object mode** (a tight Python/numpy loop, no AST walk). **Numba is required only for numeric mode**; object mode runs with numpy alone. `once` (0.4.4) lowers to a persistent `__once_fired_N` flag; UDF-scoped `once` forces object mode.



The public façade is `pynescrypt.compiler.engine` (`transpile`, `compile_script`, `run_script`), also reachable as `Runtime.run(..., mode="compile")` or `mode="auto"`.

This is an MVP with growing surface—not a full substitute for the interpreter on every script (notably multi-asset `request.*` and full strategy analytics). **0.3.16–0.3.17** recovered a large object-mode residual class: UDF locals, `nopython` `None` /unicode helpers, matrix handles, UDT returns, statement `switch` assign, drawing/ `chart.point` copy, `round()` extras, import stubs as `na`, color type dispatch, and nested UDT method fields. Official TV builtins are **0 MISMATCH** vs interpret (**P1p** closed on that set); optional set0x community corpus is unmeasured, not a missing language surface.

Conceptual model

```
flowchart LR
    Pine[Pine source] --> Parse[parse AST]
    Parse --> CV[CompilerVisitor]
    CV -->|numeric subset| Num["@numba.njit loop"]
    CV -->|UDT/map/label/strategy| Obj[Object-mode loop]
    Num --> Exec[execute_script_compiled]
    Obj --> Exec
    OHLCV[OHLCV + time arrays] --> Exec
    Exec --> Out[plots / drawings / events]
```

Generated entry point is always:

```
execute_script_compiled(open_arr, high_arr, low_arr, close_arr, vol_arr, time_arr)
```

Host packing lives in `engine.CompiledScript.run` (and `pynescrypt.runtime.Runtime._run_compiled`).

Interface surface

```
from pynescrypt.compiler.engine import transpile, compile_script, run_script, has_numba

src = transpile(pine_text)          # generated Python string
cs = compile_script(pine_text)      # CompiledScript
result = cs.run(open, high, low, close, volume, time=None)
# result: dict of plot title → float64 array, optional __drawings, __events, ...
```



API	Role
<code>transpile</code>	Parse + visit → source string (inspect/debug)
<code>compile_script</code>	<code>exec</code> generated code, warm-up call, return <code>CompiledScript</code>
<code>run_script</code>	One-shot compile+run (prefer cache <code>CompiledScript</code> for batches)
<code>has_numba</code>	Whether numeric (njit) mode can load
<code>prewarm_numba_builtins</code> / <code>prewarm_scripts</code>	Host cold-start (H2 warm path)
<code>compile_cache_stats</code> / <code>compile_deploy_config</code>	Cache + deploy diagnostics
<code>clear_compile_cache</code>	In-process LRU only
<code>clear_disk_compile_cache</code>	Disk IR / index under cache dir
<code>clear_numba_function_caches</code>	Purge Numba <code>.nbi</code> / <code>.nbc</code> under known dirs

`CompiledScript` fields: `source`, `generated_code`, `execute`, `plot_titles`, `plot_kinds` (per-plot `plot` / `hline` / `fill` tags used to synthesize numeric-mode `__drawings`), `object_mode`, optional `nopython_fallback_reason`.

`CompiledScript.run` signature

```
def run(
    self,
    open_, high, low, close,
    volume=None, # default: ones
    time=None,   # bar-open Unix ms; optional 6th series
) -> dict[str, Any]:
```

Argument	Behavior
OHLCV	Coerced to contiguous <code>float64</code> ; lengths must match (<code>ValueError</code> otherwise)
<code>volume=None</code>	Filled with ones
<code>time=None</code>	Synthetic <code>bar_index * 60_000</code> ms (unit-test / pure-compile default)
<code>time=arr</code>	Real bar-open timestamps; length must match OHLCV

`Runtime._run_compiled` packs OHLCV dicts and **passes real bar times** so calendar/ `time` / `timestamp` / `time[n]` stay aligned with interpret. Pure `cs.run(o,h,l,c)` without `time=` is fine for pure-price scripts and fixtures.

When `time` is omitted, the engine synthesizes `np.arange(n) * 60000.0` (1-minute bar opens from epoch). Hosts that care about `year` / `month` / `time[n]` must pass bar-open Unix ms.

Warm compile (product defaults)

Pro API `/run` defaults to `mode=auto` (compile when eligible, interpret on fallback with `compile_fallback_reason`). Disk IR cache is `on` (`PYNE_COMPILE_DISK_CACHE=1`); Docker sets `PYNE_COMPILE_CACHE_DIR=/data/compile-cache`. Operators call `POST /compile/prewarm` or `pynscript prewarm` so first interactive latency skips Numba cold JIT. Design notes live in repo `docs/COMPILER_PLAN.md` (not a product page).

`Runtime._compile_eligible` (used by `mode=auto`) **skips compile** when the source contains a top-level `import` or any `request.` token — those stay interpret even though the emitter can lower same-symbol `request.security`. Explicit `mode=compile` still attempts the visitor.

Mode selection

`CompilerVisitor` sets `object_mode = True` when it sees:

- User-defined types / enums
- Map / array / matrix handles that cannot stay float64
- Drawing constructors (`label.new` , `line.new` , ...) and assigned `hline` handles
- Strategy usage (object mode + `CompileStrategyBroker`)
- String/color series, library imports, or other non-numeric constructs

Statement-form `hline(price)` / `fill(...)` with a proven-numeric price **stay nopython**: they emit a constant (or all- `nan`) series plus `plot_kinds` metadata; the engine synthesizes `__drawings` after the JIT loop. Mixed scripts that later flip object mode replay those events into the body.

Otherwise **numeric mode** is preferred. Numeric mode **requires Numba** at load/warm-up; missing Numba raises `CompileNumbaRequiredError` . Object mode does **not** need Numba (still star-imports `numba_builtins` for `safe_*` / non-jitted helpers).

On numeric warm-up **TypeError** / `nopython` failure, the engine re-emits with `force_object_mode=True` and records `nopython_fallback_reason` —so `mode="compile"` can still run without hard-failing on partial numeric coverage.

Numba is **not** a hard requirement for all compile. Object-mode scripts (strategy, drawings, UDT, maps) execute without it. Only a pure-numeric emit that never flips to object mode needs Numba installed.

What numeric mode lowers today

Series assigns, arithmetic/logic, history (`close[n]`), `if` / `for` / `while` , selected `ta.*` (and bare TA aliases), scalar math helpers, `plot` , `input.*` defaults, `var` / `varip` carry—executed under `@numba.njit(cache=False)` on the generated entry (`cache=False` because code is `exec` 'd from a string without a stable disk locator). Disk-cached IR modules may rewrite to `cache=True` for cross-process reuse of machine code.

Object mode extras

- UDT instances as field dicts
- Maps as Python dicts
- `__drawings` list of structured events
- Optional `__strategy` broker: pending fills, position, `__events` , equity snapshots



Compile-path limits (notable)

Construct	Compile behavior
<code>request.security</code> / bare <code>security</code> / <code>request.security_lower_tf</code> / <code>request.seed</code>	Same-symbol, simple OHLCV expression only (chart identity + bare <code>close</code> / <code>high</code> / ... style). Foreign symbol or complex expression → <code>np.nan</code> (no inventing chart close as foreign data). Other <code>request.*</code> → object-mode <code>np.nan</code> stub
Bare builtins vs user series	User-defined series arrays win in <code>visit_Name : ad = ta.cum(...)</code> / <code>tr = ...</code> shadow bare <code>ad</code> / <code>tr</code> formulas so <code>plot(ad)</code> is not silently rebound to the builtin
<code>hline(...)</code>	Statement-form + numeric price → nopython constant series (<code>hline</code> , <code>hline_2</code> , ...) + synthesized <code>__drawings</code> . Assigned / non-numeric handles → object mode
<code>fill(..., title=...)</code>	Statement-form → nopython all- <code>nan</code> series key (band color in <code>plot_meta</code> / <code>drawings</code>). Expression-form / mixed object scripts emit <code>__drawings</code> in-loop

Compile-path `request.security` never invents foreign OHLCV from chart close. Foreign tickers and complex expressions (UDFs, `ta.*` inside the third arg, multi-value constructs beyond simple chart series) lower to `np.nan` . Use interpret + a real `data_feed` / `data_provider` for multi-asset work.

Internals

Path	Role
<code>src/pynascript/compiler/compiler.py</code>	<code>CompilerVisitor</code> , emit numeric/object, security/name/fill/hline lowering
<code>src/pynascript/compiler/numba_builtins.py</code>	JIT kernels + object-mode <code>safe_*</code>
<code>src/pynascript/compiler/strategy_broker.py</code>	Compile broker
<code>src/pynascript/compiler/engine.py</code>	Façade, caches, <code>CompiledScript.run</code> , Numba cache recovery
<code>src/pynascript/runtime/host.py</code>	Package SoT <code>Runtime._run_compiled</code> envelope; packs OHLCV + <code>time=</code>
<code>backend/runtime.py</code>	Compat re-export of <code>pynascript.runtime.host</code>
<code>scripts/compare_interp_compile.py</code>	Interpret ↔ compile series parity harness
<code>tests/test_compiler_numba.py</code> , <code>test_compiler_objects.py</code> , <code>test_compiler_strategy.py</code>	Coverage
<code>docs/COMPILER_PLAN.md</code>	Design history and remaining work (repo, not hosted)

Data layout

Unlike interpreter `PineSeries` dequeues:

```

open_arr, high_arr, low_arr, close_arr, vol_arr, time_arr
user_arr = np.full(n_bars, np.nan)          # or dtype=object for UDT
plot_i    = np.full(n_bars, np.nan)
for __bar_idx in range(n_bars):
    user_arr[__bar_idx] = ...
    plot_i[__bar_idx] = ...

```

- Bare `time` / `time_close` / `last_bar_time` / calendar extractors read `time_arr` (synthetic when omitted).
- `var` lowers to “init only when still na / first write” patterns so carry matches Pine without declaration sets.
- Numeric mode returns a **tuple** of plot arrays (host maps titles); object mode returns a **dict** (plots + `__drawings` / strategy extras).

Caches and corrupt-Numba recovery

Three layers:

1. **In-process source LRU** (sha256 of raw and/or sanitized source) + secondary IR cache keyed by generated-code hash.
2. **Disk IR** (default on): modules under `PYNE_COMPILE_CACHE_DIR` or `$XDG_CACHE_HOME/pynscript/compile`. The source → IR index JSON carries `"v": 9` (`engine._DISK_META_VERSION`). Bump that integer when generated IR semantics change so stale modules are ignored (source hash alone is stable across emitter fixes).
3. **Numba function cache** (`.nbi` / `.nbc`) next to `numba_builtins` and under disk-IR `__pycache__`.

Truncated/corrupt Numba pickle files raise `EOFError` / `pickle.UnpicklingError` on load. The engine wraps execute/warm paths with **purge + single retry** via `clear_numba_function_caches` (does not fail the script for a bad cache alone). Manual clean slate:

```

from pynscript.compiler import (
    clear_compile_cache,
    clear_disk_compile_cache,
    clear_numba_function_caches,
)
clear_compile_cache()
clear_disk_compile_cache()
clear_numba_function_caches()
# shell: rm -rf ~/.cache/pynscript/compile \
#   src/pynscript/compiler/__pycache__/numba_builtins*.nb*

```

Clear disk + Numba caches after compiler emitter or kernel edits if results look stale.

Invariants & edge cases

1. **Same AST front-end**. No second parser—compile bugs are lowering bugs.
2. **OHLCV (+ time when provided) length equality** enforced in `CompiledScript.run`.
3. **Warm-up** ignores exceptions on a dummy 16-bar series (JIT or first-run); non-nopython failures may surface on the first real run.
4. **Result normalization** converts Numba typed maps to plain dicts; `__drawings` / `__events` stay Python lists.
5. **User series shadow bare builtins** in `visit_Name` (`ad`, `tr`, `n`, ...) once `*_arr` is allocated.
6. **request.security** is not multi-asset on compile. Only chart-symbol simple OHLCV passthrough; everything else is `na`.

7. **Interpret still owns the full surface.** Exotic builtins, real multi-symbol feeds, and rich strategy metrics remain interpret-first until lowered.

Worked example

Pine:

```
//@version=6
indicator("c")
s = ta.sma(close, 14)
plot(s, "sma")
```

Generated shape (illustrative; numeric mode returns a tuple of series):

```
@numba.njit(cache=False)
def execute_script_compiled(open_arr, high_arr, low_arr, close_arr, vol_arr, time_arr):
    nBars = len(close_arr)
    s_arr = np.full(nBars, np.nan)
    plot_0 = np.full(nBars, np.nan)
    for __bar_idx in range(nBars):
        s_arr[__bar_idx] = numba_sma(close_arr, 14, __bar_idx)
        plot_0[__bar_idx] = s_arr[__bar_idx]
    return (plot_0,)
# Host packs titles → {'sma': plot_0}
```

Interpret ↔ compile parity

Full contract (tolerances, buckets, intentional diffs): [Interpret ↔ compile parity](#).

```
# From repo root – nan-aware allclose on common series keys
python scripts/compare_interp_compile.py --bars 1000 --limit 50
python scripts/compare_interp_compile.py --glob 'ta_*.pine' --bars 200
python scripts/compare_interp_compile.py --files tests/fixtures/parity/pine/strategy_01_entry_long.
python scripts/compare_interp_compile.py --ignore-hline-keys --ignore-fill-keys --strict-keys
```

Writes `.cache/interp_compile_parity.json`. First-party `hline` / `fill` / `bgcolor` / `plotshape` / `barcolor` / `plotarrow` / `plotbar` / `plotcandle` keys match interpret ↔ compile (0.3.12); `drawings` is geometry-only. Ignore flags are for leftover corpus noise.



Failure modes

Symptom	Cause
<code>numba is required for numeric compile mode</code> (<code>CompileNumbaRequiredError</code>)	Install numba or force/land in object mode (drawings/UDT/strategy already do)
Empty generated code	Visitor returned blank—unsupported top-level shape (<code>CompileEmitError</code>)
Missing <code>execute_script_compiled</code>	Emit bug or failed <code>exec</code> (<code>CompileLoadError</code>)
<code>EOFError</code> / <code>UnpicklingError</code> during JIT load	Corrupt Numba <code>.nb*</code> —engine purges and retries; if stuck, call <code>clear_numba_function_caches</code>
Silent semantic drift	Kernel seed differences—run <code>scripts/compare_interp_compile.py</code>
All- <code>nan</code> from <code>request.security</code>	Foreign symbol or non-simple expression on compile path (by design)
Wrong values for <code>ad</code> / <code>tr</code> after assignment	Fixed by user-series shadowing; if stale, clear compile/Numba caches
Strategy metrics missing	Use interpret or inspect <code>__events</code> / broker fields only
Calendar/ <code>time</code> off by hours	Host omitted <code>time=</code> ; synthetic 1m bar opens used

Remaining work (summary)

Expand Numba `ta.*` surface, richer drawings, nested UDT methods, broader `request.*` (still interpret-first for multi-asset; `mode=auto` already skips any `request.` source), and grow systematic interpret ↔ compile parity coverage per lowered construct. In-process + disk IR cache (`_DISK_META_VERSION = 9`) and product prewarm are landed. See repo `docs/COMPILER_PLAN.md` for the living checklist.

See also

- [Numba path](#)
- [Strategy broker](#)
- [Interpret ↔ compile parity](#)
- [Runtime hub](#)
- [request.* / input.* \(interpret\)](#)
- [Numerical validation](#)

NUMBA

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title: "Numba path" description: "JIT bar loops, numba_builtins kernels, NA-as-NaN, and performance characteristics."

NUMBA PATH

Abstract

Numeric compile mode emits a single `@numba.njit` function whose body is the script's bar loop. Pine builtins that cannot cross the JIT boundary are replaced by **hand-written kernels** in `numba_builtins.py` that take full arrays plus the current index (or a small float state vector for `*_inc` forms). The result is large speedups on multi-thousand-bar series after a one-time JIT warm-up—at the cost of a restricted language subset and `np.nan` as the missing-value sentinel.



Kernel contracts below match interpret `ta.*` / calendar helpers where parity was fixed; intentional deviations should be documented on the kernel docstring and here.

Conceptual model

```
flowchart TB
    CV[CompilerVisitor] --> Emit["@numba.njit execute_script_compiled"]
    Emit --> NB[numba_builtins]
    NB --> SMA[numba_sma / sma_inc]
    NB --> EMA[numba_ema_inc / rma_inc]
    NB --> RSI[numba_rsi / rsi_inc]
    NB --> HL[numba_highest / lowest / highestbars]
    NB --> ROC[numba_roc]
    NB --> Cal[numba_timestamp / utc_parts]
    NB --> Math[numba_nz / abs / min / max]
    Emit --> Loop["for __bar_idx in range(n)"]
    Loop --> SMA
    Loop --> EMA
    Loop --> RSI
    Loop --> HL
    Loop --> ROC
    Loop --> Cal
    time_arr[time_arr] --> Cal
```

Interface surface

Kernels are defined with `@numba.njit(cache=True)` in `numba_builtins.py`. Generated entry points use `@numba.njit(cache=False)` in the in-memory IR string; when the engine writes a **disk IR** module it rewrites that decorator to `cache=True` so Numba can store machine code next to a real file path (see [Cache](#)).

Generated code imports `from pynescrypt.compiler.numba_builtins import *` so call sites are bare names. Prefer `*_inc` when the visitor allocates fixed state (`__ema0_st, ...`).



Kernel table (interpret-parity highlights)

Kernel	Semantics
<code>numba_sma(arr, period, i)</code>	Mean of last <code>period</code> bars ending at <code>i</code> ; <code>nan</code> if warm-up or any window <code>nan</code>
<code>numba_ema</code> / <code>numba_ema_inc</code>	SMA seed, then EMA ($\alpha = 2/(period+1)$). Inc seed is NaN-safe : while the accumulator is still <code>nan</code> and <code>j >= period-1</code> , try SMA over <code>arr[j-period+1 : j+1]</code> only when every sample is finite. Nested EMA / DEMA no longer stuck all-NaN after leading NaNs
<code>numba_rma_inc(arr, period, i, st)</code>	Wilder RMA with the same NaN-window-safe SMA seed as <code>numba_ema_inc</code> . After seed, NaN inputs hold the previous RMA. Required for <code>rma(tr)</code> / expanded ADX so compile is not all-NaN or ~0 after warm-up
<code>numba_rsi</code> / <code>numba_rsi_inc</code>	Wilder RSI , not rolling simple-window RSI. SMA of first <code>period</code> deltas (bars <code>1..period</code>), then RMA of gain/loss ($\alpha = 1/period$). First valid bar is <code>i == period</code> (<code>period</code> deltas need <code>period+1</code> prices). <code>avg_loss == 0</code> → <code>100.0</code>
<code>numba_highest</code> / <code>numba_lowest</code>	Window extrema; full window required (<code>i >= period-1</code>); NaN samples skipped
<code>numba_highestbars</code> / <code>numba_lowestbars (+ _inc)</code>	TV negative bars-back offsets: <code>0</code> if current is extreme, <code>-1</code> one bar ago, ..., <code>-(length-1)</code> . Returns <code>-1.0</code> when window not full, invalid length, or all-NaN. Oldest extreme wins ties (strict <code>></code> / <code><</code>)
<code>numba_roc(arr, length, i)</code>	$100 * (src - src[length]) / src[length]$ at bar <code>i</code> ; <code>nan</code> if <code>i < length</code> , baseline 0/NaN, or <code>src</code> NaN
<code>numba_timestamp(y, m, d, h=0, mi=0, s=0)</code>	UTC epoch ms from calendar components; month/day overflow normalized (e.g. <code>month=0</code> → prior Dec) for TTM-style windows
<code>numba_utc_parts(ms)</code>	(<code>year, month, dayofmonth, hour, minute, second, dayofweek</code>) floats; Pine <code>dayofweek</code> is <code>1=Sun ... 7=Sat</code>
<code>numba_nz(val, replacement)</code>	Replace <code>nan</code>
<code>numba_abs</code> / <code>numba_min</code> / <code>numba_max</code>	Scalar helpers

Full inventory lives in `src/pynescrypt/compiler/numba_builtins.py` (njit batch + `*_inc` + pure-Python object-mode helpers).

Calendar / `time_arr`

Bar open time is the host `time_arr` series (synthetic when the runner omits timestamps).

Pine form	Lowering
<code>bare year / month / dayofmonth / hour / minute / second / dayofweek</code>	<code>numba_utc_parts(time_arr[__bar_idx])[idx]</code>
<code>year(t)</code> ... (optional time arg)	<code>numba_utc_parts(src)[idx]</code> with <code>src</code> defaulting to <code>time_arr[__bar_idx]</code>
<code>timestamp(y, m, d[, h, mi, s])</code>	<code>numba_timestamp(...)</code> (leading timezone string args dropped → UTC stub)
<code>time</code> / <code>time_close</code> (approx)	<code>time_arr[__bar_idx] / + 59999 ms</code>



History access

`close[n]` lowers to array indexing at `__bar_idx - n` with bounds $\rightarrow$ `nan` (parallel to interpreter `None`, different sentinel).

Control flow

`if` / `for` / `while` / `once` emit Python control flow **inside** the njit function—supported Numba subset only (no heap objects, no Python lists of mixed types). `once` lowers to a persistent `__once_fired_N` bool set when the body runs (compile is historical: `barstate.isconfirmed` is always true). A `once` inside a UDF forces object mode so the flag is module-level.

Cache

Three layers interact:

Layer	Location / key	Role
Source LRU	in-process, sha256 of source	<code>CompiledScript</code> reuse
IR LRU	in-process, sha256 of generated Python	share warm callable across comment-only variants
Disk IR	<code>PYNE_COMPILE_CACHE_DIR</code> or <code>\$XDG_CACHE_HOME/pynescrypt/compile</code>	persist generated modules; disable with <code>PYNE_COMPILE_DISK_CACHE=0</code> . Index JSON <code>"v"</code> must equal <code>engine._DISK_META_VERSION</code> (9)

Numba function cache (`.nbi` / `.nbc`):

- Hand-written kernels: `@njit(cache=True)` $\rightarrow$ artifacts under `.../pynescrypt/compiler/__pycache__`.
- Generated entry: IR is emitted with `cache=False` (safe for `exec` from `<string>`). Disk write rewrites `@numba.njit(cache=False)` $\rightarrow$ `cache=True` so Numba's file locator can cache machine code under the disk IR `__pycache__`.
- Truncated/corrupt pickle loads raise `EOFError` / `UnpicklingError`. The engine purges known `.nbi` / `.nbc` via `clear_numba_function_caches()` and recompiles once instead of failing the script.

```
from pynescrypt.compiler import clear_numba_function_caches, clear_disk_compile_cache
clear_numba_function_caches()
clear_disk_compile_cache()
```

Internals

Path	Role
<code>src/pynescrypt/compiler/numba_builtins.py</code>	Kernels (<code>cache=True</code>)
<code>src/pynescrypt/compiler/compiler.py</code>	<code>_emit_numeric_mode</code> , call lowering, <code>time_arr</code> calendar
<code>src/pynescrypt/compiler/engine.py</code>	Disk IR rewrite, Numba cache recovery, prewarm
<code>tests/test_compiler_numba.py</code>	Correctness / parity

Expanding coverage is primarily **new kernels + visitor call mapping**, not changes to the bar-loop skeleton.



Invariants & edge cases

1. **No Python objects in numeric mode.** Strings, UDTs, maps, drawings force object mode before njit is applied.
2. **Warm-up cost.** First `compile_script` invokes a dummy run; production servers should cache `CompiledScript` and prewarm (`PYNE_COMPILE_PREWARM` default `on`, `POST /compile/prewarm`).
3. **EMA/RMA seed** uses sliding all-finite SMA windows—not “seed once at bar `period-1` only if that fixed slice is clean.” Leading NaN on `tr` / nested EMA sources must eventually produce values.
4. **RSI is Wilder**, first valid at `i == period` . Do not compare against a simple N-bar average-gain RSI.
5. **highestbars/lowestbars** return negative offsets and `-1` on incomplete windows; ties keep the oldest bar.
6. **Disk entry cache** requires the rewrite to a real file path; pure `exec` of `cache=False` IR does not populate Numba’s on-disk function cache for the entry function.
7. **Anaconda static libpython.** Packaging notes in AGENTS/build docs apply when shipping JIT-heavy binaries.

Worked examples

Benchmark mental model

```
interpret: 0(bars × ast_nodes × python_dispatch)
numeric:   0(bars × lowered_ops) in machine code after JIT
```

For simple SMA scripts, internal notes cite order-of-magnitude speedups versus list-based interpretation on long series (see repo `docs/COMPILER_PLAN.md` qualitative claims).

Using from Pro API

```
Runtime(...).run(source, ohlcv, mode="compile")
```

Converts bar dicts → arrays (including `time=` bar-open ms), compiles, reshapes plots into the standard envelope, flags `object_mode` when applicable. `generated_code` is omitted unless `PYNESCRIPT_RETURN_GENERATED_CODE=1` .



Failure modes

Symptom	Cause
TypeError from Numba	Unsupported construct leaked into numeric emit
All- <code>nan</code> plots	Period longer than series; history index bugs; or (fixed) NaN-poisoned EMA/RMA seed on nested sources
Nested DEMA / <code>rma(tr)</code> all- NaN	Pre-fix seed; current kernels sliding-window all-finite SMA seed
RSI diverges vs interpret	Expect Wilder seed/RMA, not rolling window RSI
<code>highestbars</code> sign / magnitude off	TV negative offsets; incomplete window → <code>-1</code>
<code>EOFError</code> / <code>UnpicklingError</code> at load	Corrupt <code>.nbi</code> / <code>.nbc</code> — engine purges and retries; manual <code>clear_numba_function_caches()</code>
ImportError numba	Environment missing dependency
Object mode unexpectedly	Visitor detected drawing/UDT/map—inspect <code>visitor.object_mode</code>

Performance notes

- Prefer **one compile, many runs** with different OHLCV.
- Keep scripts in the numeric subset for realtime ticks.
- Host prewarm of `numba_builtins` + common scripts shifts cold JIT off the first user request.
- Object mode still wins over AST walking for UDT-heavy scripts but will not match peak njit throughput.

See also

- [Compiler overview](#)
- [Technical builtins \(interpret\)](#)
- [Series & history](#)

STRATEGY BROKER

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title: "Compile strategy broker" description: "CompileStrategyBroker: pending orders, OHLC fills, OCA, commission, and event export."

COMPILE STRATEGY BROKER

Abstract

When `CompilerVisitor` detects strategy usage, object-mode emission instantiates `CompileStrategyBroker` —a lightweight broker aligned with the interpreter's fill-before-script cadence, but designed for generated loops (no AST, no full `StrategyState` metric surface). It supports market entry/close and pending limit/stop/stop-limit orders with per-bar OHLC triggers, OCA cancel/reduce, commission, and slippage.

Conceptual model

```
sequenceDiagram
    participant Loop as Generated for __bar_idx
    participant B as CompileStrategyBroker
    Loop->>B: begin_bar(idx, OHLC, time)
    Note over B: set_bar + process_pending when book nonempty
    Loop->>B: entry/close/order... (script body)
    Note over B: events[], pending_orders, position_*
    Loop->>B: to_events() at return
```

Interface surface

Construction (from `strategy()` kwargs when lowered)

```
CompileStrategyBroker(
    initial_capital=100_000.0,
    commission_value=0.0,
    commission_type="percent", # percent | cash_per_order | cash_per_contract
    slippage_ticks=0,
    mintick=0.01,
    pyramiding=0,
    default_qty_type="fixed",
    default_qty_value=1.0,
    avg_price_model="stock", # stock | futures | inverse (pynescrypt extension)
    leverage=1.0,           # futures UI: buying power / margin divisor
)
```

`avg_price_model` is lowered from `strategy(..., avg_price_model=...)`. The compile broker is a **single net lot**: partial reduce already keeps `position_avg_price` sticky (futures-like). Multi-leg stock reweight on partial close is interpret-first until the compile path grows an open-trade list.

`leverage` scales percent/cash default qty ($\text{margin} \times \text{leverage} / \text{price}$) and free `cash` (equity – notional/leverage).

Bar context

```
# Preferred generated hot path (one call; skips pending walk when empty)
broker.begin_bar(bar_index, open_, high, low, close, bar_time=...)
# Equivalent two-step still exists:
broker.set_bar(bar_index, bar_time, mark, open_=..., high=..., low=..., close=...)
broker.process_pending_orders(open_=..., high=..., low=..., close=...)
```

Orders

Method	Role
<code>entry(...)</code>	Market or pending entry; reverse flattens opposite
<code>close</code> / <code>close_all</code>	Reduce/flat with optional price
Pending book	<code>PendingOrder</code> with type, direction, qty, limit/stop, OCA, <code>max_fill_per_bar</code> , <code>is_entry</code>



Fill logic (`_trigger_price`)

Type	Long trigger	Short trigger	Fill price idea
market	—	—	close (immediate path)
limit	<code>low <= limit</code>	<code>high >= limit</code>	gap-aware vs open
stop	<code>high >= stop</code>	<code>low <= stop</code>	gap-aware vs open
stop-limit	stop traded and limit available	symmetric	limit price

Partial fills: `max_fill_per_bar` caps quantity per bar; residual stays pending.

OCA

After a fill, siblings in the same `oca_name` :

- **cancel** — remove sibling, emit `cancel`
- **reduce** — decrease sibling qty by fill size
- **none** — no interaction

Economics

- **Slippage**: `± slippage_ticks * mintick` by direction on fills.
- **Commission**: percent of notional, cash per order, or cash per contract.
- **Position**: signed `position_size`, `position_avg_price`, `netprofit`, `equity` helpers for return payload.

Events

`_emit` appends dicts with keys compatible with strategy event serialization (`kind`, `id`, `direction`, `qty`, `order_type`, `limit`, `stop`, `oca_name`, `comment`, `bar_index`, `bar_time`, `ohlcv`). Object-mode return includes `'__events': __strategy.to_events()` plus position/netprofit/equity snapshots.

Internals

Path	Role
<code>src/pynescrypt/compiler/strategy_broker.py</code>	Broker implementation
<code>src/pynescrypt/compiler/compiler.py</code>	Emits <code>__strategy</code> wiring in object mode
<code>src/pynescrypt/ast/evaluator/builtins/strategy.py</code>	Interpreter counterpart (richer)
<code>tests/test_compiler_strategy.py</code>	Compile strategy coverage

NA helpers treat `None`, blank/ `na` strings, and float NaN as missing prices when classifying order types.

Invariants & edge cases

1. **Same order as interpreter**: `begin_bar` (or `set_bar` → `process_pending_orders`) → script body.
2. **Closed-trade list is partial**. The compile broker now keeps `closed_trade_records` and some `closedtrades_*` accessors (profit, comments, max drawdown/runup). Prefer interpret for the full `strategy.closedtrades.*` / `opentrades.*` metric surface.
3. **Direction normalization** accepts `strategy.long`, `long`, `1`, `buy`, etc.



- 4. **Reverse on opposite entry** via `close_all(comment="reverse")` then open.
- 5. **Event dicts are mutable lists**—fine for run output; not frozen dataclasses.

Worked examples

Generated prologue (illustrative)

```
__strategy = CompileStrategyBroker(initial_capital=100000.0)
for __bar_idx in range(n_bars):
    __strategy.begin_bar(__bar_idx,
        float(open_arr[__bar_idx]), float(high_arr[__bar_idx]),
        float(low_arr[__bar_idx]), float(close_arr[__bar_idx]),
        bar_time=int(time_arr[__bar_idx]))
    # lowered strategy.entry / close calls
return {..., '__events': __strategy.to_events(),
    '__position_size': __strategy.position_size,
    '__netprofit': __strategy.netprofit,
    '__equity': __strategy.equity}
```

Limit buy pending

```
strategy.entry("L", strategy.long, limit=low - syminfo.mintick)
```

Object-mode lowers to a pending limit; subsequent bars fill when `low` trades through.

Failure modes

Symptom	Cause
No fills	Triggers never met; or market path not invoked
OCA sibling remains	<code>oca_type</code> none / name mismatch
PnL vs interpret drift	Commission/slippage/mintick defaults differ
Missing events in API envelope	Host only maps plots—not <code>__events</code>

See also

- [Strategy builtins \(interpret\)](#)
- [Events](#)
- [Compiler overview](#)
- [Numba path](#)

PARITY

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title: "Interpret ↔ compile parity" description: "Plot-series parity between Runtime interpret and compile modes: harness, tolerances, and known sentinel differences."

INTERPRET ↔ COMPILE PARITY

Abstract

PYNE ships two bar engines that should agree on **numeric plot series** for scripts in the compile surface: the AST interpreter and the generated Numba/object-mode loop. This page is the short contract for that agreement—how to measure it, what “equal” means, and where intentional differences live (`None` vs `nan`, host `time_arr`, structural keys).

Strategy **event** parity (Python vs TypeScript worker) is a separate oracle (`tests/test_parity.py` + fixtures). Here the focus is **plot values** on one host: `pynescrypt.runtime.Runtime`.



Conceptual model

```
flowchart LR
    Pine[*pine] --> I[Runtime mode=interpret]
    Pine --> C[Runtime mode=compile]
    I --> SI[series dict]
    C --> SC[series dict]
    SI --> Cmp[nan-aware allclose]
    SC --> Cmp
    Cmp --> Report[.cache/interp_compile_parity.json]
```

Both modes receive the **same** synthetic (or host) OHLCV. Compile additionally builds `time_arr` from bar `time` fields so `time` / calendar expressions line up with the interpret `time` series.

Interface surface

Corpus harness

```
# From repo root – first-party fixtures by default
python scripts/compare_interp_compile.py --bars 1000 --limit 50
python scripts/compare_interp_compile.py --files tests/fixtures/parity/pine/strategy_01_entry_long.
```

Flag / output	Role
Default inputs	<code>tests/fixtures/parity/pine/*.pine</code> (first-party)
Tolerances	<code>rtol=1e-5</code> , <code>atol=1e-6</code> (nan-aware)
Report	<code>.cache/interp_compile_parity.json</code>
Exit 0	No value/ <code>nan</code> mismatches on common series keys

Buckets: `OK`, `fill_background_only`, `both_error_same`, `expected_error`, `both_error`, `MISMATCH`, `interp_error`, `compile_error`, `structural_only`. Matched errors on both backends (`both_error_same`) and intentional demo failures (`expected_error`, e.g. `auto-fib` / `pivot-depth`) count as success unless `--strict-errors`. First-party `hline` / `fill` / `bgcolor` / `plotshape` / `barcolor` / `plotarrow` / `plotbar` / `plotcandle` keys match interpret ↔ compile (0.3.12); `drawings` is geometry-only. `--ignore-hline-keys` / `--ignore-fill-keys` remain optional for leftover **corpus** key noise; `--strict-keys` fails on any one-sided key.

Focused unit tests

Test	Scope
<code>tests/test_dividend_yield_parity.py</code>	Host-series copy-on-assign + time history (interpret/compile)
<code>tests/test_compiler_numba.py</code>	Kernel-level numeric correctness
<code>tests/test_parity.py</code>	Strategy event fixtures (not plot <code>allclose</code>)

Product path

Pro API `/run` defaults to `mode=auto`: compile when `_compile_eligible` passes (no top-level `import`, no `request.token`), otherwise interpret with `compile_fallback_reason`. Parity harness always forces explicit `interpret` vs `compile` so fallbacks do not hide drift.



Official TradingView® builtins (0.6.2): interpret UDF series-parameter history (`s[1]` per call site) closed the remaining P1p plot MISMATCH tail on that set — **0 MISMATCH** vs compile. Optional set0x community corpus is unmeasured, not a missing language surface.

What must match

1. **Common plot keys** — float series after `None` ↔ `nan` normalization.
2. **Warm-up `na`** — leading missing samples align (interpret `None` / compile `nan`).
3. **`time` / bar-open ms** — when OHLCV includes `time`, or both use the synthetic `i * 60_000` fallback.
4. **Structured errors** — scripts that `runtime.error` (e.g. insufficient pivots) should fail on both with messages that normalize equal (`both_error_same`).

Intentional / soft differences

Topic	Policy
Sentinels	Interpret: <code>None</code> ; Numba: <code>np.nan</code> . Compare with nan-aware equality, not <code>==</code> .
Hline / fill / bgcolor keys	May appear on one mode only; ignore with harness flags unless you need strict key sets.
Unresolved imports / stubs	Non-numeric plot cells serialize as <code>na</code> (null)—not string stubs.
Outside compile surface	UDT-heavy / full <code>request.*</code> / rich strategy analytics stay interpret-first; <code>auto</code> falls back.
Disk / Numba caches	Stale IR after kernel edits can fake mismatches—clear compile + Numba caches (see harness docstring).

Internals

Path	Role
<code>scripts/compare_interp_compile.py</code>	End-to-end series compare
<code>tests/test_interp_compile_parity.py</code>	Always-on smoke + harness flag tests
<code>src/pynscript/runtime/host.py</code>	<code>_run_compiled</code> , OHLCV + <code>time_arr</code> packing
<code>backend/runtime.py</code>	Compat re-export of the package host
<code>src/pynscript/compiler/engine.py</code>	Compile façade + result normalize
<code>src/pynscript/compiler/numba_builtins.py</code>	JIT kernels under test

Worked example

```
python scripts/compare_interp_compile.py \  
  --files tests/fixtures/parity/pine/strategy_01_entry_long.pine \  
  --bars 500  
# → OK (or MISMATCH with first differing bar/key in the JSON report)
```



Failure modes

Symptom	Likely cause
Systematic offset after warm-up	EMA/RSI seed or window clamp drift in a kernel
Only-compile empty plots	Script still object-mode / plot not lowered; or cache corruption
<code>time</code> diverges	Host omitted times on one path only; check <code>time_arr</code> packing
Structural-only keys	hline/fill/background naming — use ignore flags or align collectors

See also

- [Compiler overview](#)
- [Numba path](#)
- [Series & history](#)
- [Runtime hub](#)
- [Numerical validation](#)
- [PyneTS parity](#) — TypeScript vs Python plots